

GCSE Chemistry

Exam Walkthrough Pack

Step-by-step worked exam-style questions with full explanations

Original JDSscience exam-style questions — not copied from past papers

Introduction

This Chemistry Exam Walkthrough Pack is designed to help you understand how to approach GCSE and IGCSE exam-style questions. Every question in this booklet is an original JDScience item written to match common specification skills — it is not copied from AQA, Edexcel, OCR or other past papers.

How to use this pack: read each question, cover the walkthrough, and try answering in exam conditions first. Then read what the question is asking, follow the step-by-step thinking, compare your answer with the model answer, and study the mark breakdown to see exactly how marks are awarded.

Command words tell you what type of answer is required. Section 1 explains the most common GCSE science command words. Always underline the command word before you start writing.

Showing working matters: in calculations, examiners award method marks even when the final answer is wrong. In longer answers, a clear logical sequence helps you secure every available mark.

Examiners award marks for correct science linked to the question. Keywords from the specification matter, but a correct idea in your own words can still score. Extended questions often use level-based marking: plan before you write, use paragraphs, and link ideas.

To move from Grade 4/5 to Grade 7/8/9: use precise terminology, explain mechanisms (not just names), include data or examples where asked, and check units and significant figures in calculations. For six-mark questions, a short plan and linked paragraphs beat a long unordered list.

Section 1: Command word guide

State

What you must do: Give a short, factual answer — often one word, number or phrase. No explanation is required unless the question also asks you to explain.

Common mistake: Writing a long paragraph when a single fact is enough, or giving an explanation when only a fact was asked for.

Model sentence starter: The [quantity/name] is ...

Give

What you must do: Provide the information requested. Similar to state, but may need a short phrase or named example.

Common mistake: Listing unrelated facts that do not answer the question.

Model sentence starter: One example is ... / The value is ...

Describe

What you must do: Say what happens or what something is like. Focus on observations, patterns or features. You do not need to say why.

Common mistake: Explaining causes when the command word is only describe.

Model sentence starter: First ... then ... / The pattern shows ...

Explain

What you must do: Give reasons why something happens. Link cause and effect using science ideas and key terms.

Common mistake: Describing what happens without saying why, or using everyday language instead of science.

Model sentence starter: This happens because ... which means ...

Calculate

What you must do: Work out a numerical answer. Show your working, include units, and give the final value to the required precision.

Common mistake: Missing units, no working, or using the wrong equation.

Model sentence starter: Use ... = ... / Substitute: ...

Compare

What you must do: Identify similarities and/or differences between two or more things. Refer to both sides.

Common mistake: Only describing one item, or listing features without comparing.

Model sentence starter: Both ... however ... / X has ... whereas Y ...

Evaluate

What you must do: Weigh up evidence or options, consider strengths and weaknesses, and reach a supported judgement.

Common mistake: Only listing advantages with no conclusion or balanced comment.

Model sentence starter: An advantage is ... A limitation is ... Overall ...

Suggest

What you must do: Apply your knowledge to a new situation. Your answer should be sensible and scientifically plausible.

Common mistake: Repeating textbook facts that do not fit the context given.

Model sentence starter: A possible reason is ... / This could be because ...

Justify

What you must do: Give evidence or reasons that support a choice, conclusion or statement already made.

Common mistake: Restating the claim without giving supporting science.

Model sentence starter: This is supported by ... because ...

Determine

What you must do: Find out a value or conclusion using data, a graph or a calculation. Show how you reached your answer.

Common mistake: Reading a graph incorrectly or not showing how the value was obtained.

Model sentence starter: From the graph ... / Using the data ...

Analyse

What you must do: Break information into parts, interpret patterns or trends, and explain what the data shows.

Common mistake: Describing data without interpreting it, or ignoring anomalies.

Model sentence starter: The data shows that ... / As X increases, Y ...

Section 2: Atomic structure and the periodic table

QUESTION 1

State the relative charges of a proton, neutron and electron.

QUESTION TYPE

Topic: Atomic structure and the periodic table | Command word: state | Marks: 1 | Skill tested: Atomic structure

WHAT THE QUESTION IS ASKING

Recall the three subatomic charges tested at GCSE.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Proton: +1; neutron: 0; electron: -1.

MARK BREAKDOWN

Mark 1: Proton: +1; neutron: 0; electron: -1.

COMMON MISTAKES

- Writing neutrons as -1
- Mixing up proton and electron charges

EXAMINER TIP

List particles in the order proton, neutron, electron so you do not miss one.

QUESTION 2

Give the number of protons, neutrons and electrons in an atom of carbon-12.

QUESTION TYPE

Topic: Atomic structure and the periodic table | Command word: give | Marks: 1 | Skill tested: Atomic structure

WHAT THE QUESTION IS ASKING

Use the mass number and atomic number for carbon.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

6 protons, 6 neutrons, 6 electrons.

MARK BREAKDOWN

Mark 1: 6 protons, 6 neutrons, 6 electrons.

COMMON MISTAKES

- Using 12 neutrons
- Forgetting electrons equal protons in a neutral atom

EXAMINER TIP

Neutrons = mass number – atomic number.

QUESTION 3

State what is meant by the term isotope.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** state | **Marks:** 1 | **Skill tested:** Atomic structure

WHAT THE QUESTION IS ASKING

Define isotopes in terms of atomic structure.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons.

MARK BREAKDOWN

Mark 1: Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons.

COMMON MISTAKES

- Saying isotopes have different numbers of protons
- Only mentioning mass number without protons

EXAMINER TIP

Always include 'same element' and 'different neutrons'.

QUESTION 4

Describe the plum pudding model of the atom.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** describe | **Marks:** 2 | **Skill tested:** Models of the atom

WHAT THE QUESTION IS ASKING

Outline the key features of Thomson's model.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The atom is a ball of positive charge with negative electrons embedded in it, like plums in a pudding.

MARK BREAKDOWN

Mark 1: Ball of positive charge spread through the atom
Mark 2: Negative electrons embedded in the positive charge

COMMON MISTAKES

- Calling it the nuclear model
- Forgetting that the positive charge was spread out

EXAMINER TIP

Link the name 'plum pudding' to embedded electrons.

QUESTION 5

Explain why atoms are electrically neutral.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** explain | **Marks:** 2 | **Skill tested:** Atomic structure

WHAT THE QUESTION IS ASKING

Relate proton and electron numbers to overall charge.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

An atom has equal numbers of protons (+1) and electrons (−1), so the charges cancel.

MARK BREAKDOWN

Mark 1: Equal numbers of protons and neutrons

Mark 2: No charged particles present

COMMON MISTAKES

- Only stating 'no charge' without numbers
- Mentioning neutrons as charged

EXAMINER TIP

Use both particle names and their charges.

QUESTION 6

Give two differences between Group 1 and Group 7 elements.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** give | **Marks:** 2 | **Skill tested:** Periodic table

WHAT THE QUESTION IS ASKING

Compare reactivity and electron arrangement across groups.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Group 1 metals are more reactive down the group and form 1+ ions; Group 7 non-metals are less reactive down the group and form 1− ions.

MARK BREAKDOWN

Mark 1: Group 1 reactivity decreases down the group

Mark 2: Group 7 elements are metals

COMMON MISTAKES

- Comparing periods instead of groups
- Ignoring ion formation

EXAMINER TIP

Mention reactivity trend and ion charge for both groups.

QUESTION 7

Explain how the results of the alpha particle scattering experiment led to the nuclear model.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** explain | **Marks:** 3 | **Skill tested:** Models of the atom

WHAT THE QUESTION IS ASKING

Link observations (most pass through, some deflect, few rebound) to structure.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Most alpha particles passed straight through, showing mostly empty space. Some were deflected, showing a positive nucleus. A few rebounded, showing a small, dense, positive centre.

MARK BREAKDOWN

- Mark 1: Most particles passed through → mostly empty space
- Mark 2: Deflection/rebound → small dense positive nucleus
- Mark 3: Nuclear model replaces plum pudding

COMMON MISTAKES

- Saying electrons caused deflection
- Claiming the atom is solid

EXAMINER TIP

Use the three observations in order: pass through, deflect, rebound.

QUESTION 8

Describe the electron arrangement in sodium and explain why it forms a 1+ ion.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** describe | **Marks:** 3 | **Skill tested:** Ions

WHAT THE QUESTION IS ASKING

State configuration 2,8,1 and link to losing one electron.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Sodium has electron arrangement 2,8,1. It loses one outer electron to achieve a full outer shell, forming Na⁺ with a 1+ charge.

MARK BREAKDOWN

Mark 1: Electron arrangement 2,8,1

Mark 2: Loses one electron for full outer shell

Mark 3: Forms Na⁺

COMMON MISTAKES

- Saying sodium gains an electron
- Giving wrong shell numbers

EXAMINER TIP

Always state arrangement then explain ion formation.

QUESTION 9

Explain why the modern periodic table is arranged in order of atomic (proton) number, not relative atomic mass.

QUESTION TYPE

Topic: Atomic structure and the periodic table | Command word: explain | Marks: 4 | Skill tested: Periodic table

WHAT THE QUESTION IS ASKING

Discuss isotopes and Moseley's work; contrast with Mendeleev.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Atomic number defines the element. Isotopes have the same proton number but different mass, so mass order would place isotopes incorrectly. Moseley showed properties repeat with proton number.

MARK BREAKDOWN

Mark 1: Element identity depends on proton number

Mark 2: Isotopes share proton number but differ in mass

Mark 3: Properties repeat with proton number

Mark 4: Resolves order problems e.g. Ar/K

COMMON MISTAKES

- Only mentioning Mendeleev without isotopes
- Saying mass number defines element

EXAMINER TIP

Include isotopes as the key reason mass order fails.

GRADE 7–9 EXTENSION

Link to Moseley's X-ray work for higher grades.

QUESTION 10

Describe trends in atomic radius and reactivity down Group 1.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** describe | **Marks:** 4 | **Skill tested:** Group 1

WHAT THE QUESTION IS ASKING

Explain increasing radius and increasing reactivity with shielding and distance from nucleus.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Atomic radius increases down the group as more electron shells are added. Reactivity increases because the outer electron is further from the nucleus, more shielded, and easier to lose.

MARK BREAKDOWN

- Mark 1: Radius increases — more shells
- Mark 2: Outer electron further from nucleus
- Mark 3: More shielding
- Mark 4: Easier to lose electron → more reactive

COMMON MISTAKES

- Saying reactivity decreases
- Ignoring shielding

EXAMINER TIP

Pair each trend with a reason.

QUESTION 11

A student is given samples of lithium, sodium and potassium. Describe how you could compare their reactivity with water and explain the trend observed.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** describe | **Marks:** 6 | **Skill tested:** Group 1 practical

WHAT THE QUESTION IS ASKING

Plan a fair test and explain increasing reactivity down Group 1.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Use equal-sized fresh metal pieces and the same volume and temperature of water. Place each metal on the surface and observe effervescence, movement and flame colour. Potassium reacts most vigorously, then sodium, then lithium, showing reactivity increases down Group 1 because the outer electron is more easily lost further down the group.

MARK BREAKDOWN

- Mark 1: Fair test: same size metal, same water volume/temperature

Mark 2: Observe bubbles, movement, heat, flame

Mark 3: Trend: $K > Na > Li$ reactivity

Mark 4: Reason: outer electron further/shielded

Mark 5: Easier ionisation down group

Mark 6: Safety: small pieces, goggles, dry metals

COMMON MISTAKES

- Using different water volumes
- No safety mention
- Wrong reactivity order

EXAMINER TIP

Extended responses need method, results trend and scientific explanation.

GRADE 7–9 EXTENSION

Reference ionisation energy decreasing down the group.

QUESTION 12

An atom of chlorine has 17 protons and 18 neutrons. Calculate its mass number.

QUESTION TYPE

Topic: Atomic structure and the periodic table | Command word: calculate | Marks: 2 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Add protons and neutrons to find mass number.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Mass number = protons + neutrons = $17 + 18 = 35$.

MARK BREAKDOWN

Mark 1: Correct addition of $17 + 18$

Mark 2: Mass number = 35

COMMON MISTAKES

- Using only protons
- Subtracting instead of adding

EXAMINER TIP

Mass number is nucleon number — protons plus neutrons only.

QUESTION 13

A student heats a piece of lithium in a Bunsen flame and records the flame colour. Describe the method and explain what the result shows about electron transitions.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** describe | **Marks:** 4 | **Skill tested:** Flame tests

WHAT THE QUESTION IS ASKING

Outline flame test method and link colour to electron energy levels.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Clean a wire loop, dip in lithium compound, and hold in the non-luminous Bunsen flame. A crimson-red flame is seen. Heat gives electrons energy to jump to higher levels; as they fall back, light of a characteristic frequency is emitted.

MARK BREAKDOWN

- Mark 1: Clean loop to avoid contamination
- Mark 2: Non-luminous flame
- Mark 3: Crimson-red colour for lithium
- Mark 4: Electrons absorb energy then emit light as they drop levels

COMMON MISTAKES

- Using a dirty loop
- Confusing lithium with strontium colour

EXAMINER TIP

Practical questions need clear steps and scientific explanation.

Section 2: Bonding, structure and properties

QUESTION 14

State the type of particle that carries the current in a molten ionic compound.

QUESTION TYPE

Topic: Bonding, structure and properties | Command word: state | Marks: 1 | Skill tested: Ionic bonding

WHAT THE QUESTION IS ASKING

Identify mobile charged species in molten salts.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Ions.

MARK BREAKDOWN

Mark 1: Ions.

COMMON MISTAKES

- Electrons only
- Atoms

EXAMINER TIP

Molten ionic substances conduct via mobile ions, not electrons.

QUESTION 15

Give the type of bonding in a molecule of oxygen, O₂.

QUESTION TYPE

Topic: Bonding, structure and properties | Command word: give | Marks: 1 | Skill tested: Covalent bonding

WHAT THE QUESTION IS ASKING

Distinguish intermolecular forces from bonding within a molecule.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Covalent (double covalent) bonding.

MARK BREAKDOWN

Mark 1: Covalent (double covalent) bonding.

COMMON MISTAKES

- Ionic bonding
- Metallic bonding

EXAMINER TIP

O₂ is a simple covalent molecule.

QUESTION 16

State why graphite can conduct electricity.

QUESTION TYPE

Topic: Bonding, structure and properties | Command word: state | Marks: 1 | Skill tested: Giant covalent structures

WHAT THE QUESTION IS ASKING

Link delocalised electrons to conduction in graphite.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Each carbon atom has one delocalised electron that is free to move and carry charge.

MARK BREAKDOWN

Mark 1: Each carbon atom has one delocalised electron that is free to move and carry charge.

COMMON MISTAKES

- Ions move through layers
- Metallic bonding present

EXAMINER TIP

Mention delocalised electrons specifically.

QUESTION 17

Describe the structure of an ionic compound such as sodium chloride.

QUESTION TYPE

Topic: Bonding, structure and properties | Command word: describe | Marks: 2 | Skill tested: Ionic bonding

WHAT THE QUESTION IS ASKING

Explain the giant lattice of alternating ions.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A giant ionic lattice of alternating Na⁺ and Cl⁻ ions held by strong electrostatic attractions in all directions.

MARK BREAKDOWN

- Mark 1: Giant lattice of alternating positive and negative ions
- Mark 2: Strong electrostatic forces in all directions

COMMON MISTAKES

- Calling it simple molecules
- Saying molecules of NaCl

EXAMINER TIP

Use 'giant ionic lattice' — never 'molecules' for NaCl.

QUESTION 18

Explain why diamond is hard.

QUESTION TYPE

Topic: Bonding, structure and properties | **Command word:** explain | **Marks:** 2 | **Skill tested:** Giant covalent structures

WHAT THE QUESTION IS ASKING

Link tetrahedral network and strong covalent bonds to hardness.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Each carbon is bonded to four others by strong covalent bonds in a rigid tetrahedral network, so a large force is needed to break the structure.

MARK BREAKDOWN

Mark 1: Four strong covalent bonds per carbon

Mark 2: Rigid 3D network requires large force to break

COMMON MISTAKES

- Weak intermolecular forces
- Layers slide over each other

EXAMINER TIP

Hardness comes from strong bonds throughout, not between molecules.

QUESTION 19

Give two properties of metals that are explained by metallic bonding.

QUESTION TYPE

Topic: Bonding, structure and properties | **Command word:** give | **Marks:** 2 | **Skill tested:** Metallic bonding

WHAT THE QUESTION IS ASKING

Link delocalised electrons to conduction and malleability.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Good electrical conductivity (delocalised electrons move) and malleability (layers of positive ions can slide while electrons still hold structure together).

MARK BREAKDOWN

Mark 1: Conductivity due to moving delocalised electrons

Mark 2: Malleability because layers slide without breaking metallic bond

COMMON MISTAKES

- Attributing properties to intermolecular forces

- Only naming properties without explanation

EXAMINER TIP

Always pair property with bonding explanation.

QUESTION 20

Compare the melting points of sodium chloride and chlorine. Explain your answer.

QUESTION TYPE

Topic: Bonding, structure and properties | Command word: compare | Marks: 3 | Skill tested: Structure and properties

WHAT THE QUESTION IS ASKING

Contrast giant ionic lattice with simple covalent molecules.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Sodium chloride has a high melting point because strong ionic bonds in the giant lattice need much energy to break. Chlorine has a low melting point because only weak intermolecular forces between simple molecules need overcoming.

MARK BREAKDOWN

- Mark 1: NaCl: giant ionic lattice, strong ionic bonds, high mp
Mark 2: Cl₂: simple molecules, weak intermolecular forces, low mp
Mark 3: Correct comparison of bond strengths

COMMON MISTAKES

- Saying covalent bonds break in Cl₂ melting
- Ignoring intermolecular forces

EXAMINER TIP

State structure, bond/force type, then melting point.

QUESTION 21

Describe the bonding in a polymer such as poly(ethene).

QUESTION TYPE

Topic: Bonding, structure and properties | Command word: describe | Marks: 3 | Skill tested: Polymers

WHAT THE QUESTION IS ASKING

Explain repeating unit and strong/weak forces.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A long chain of carbon atoms joined by strong covalent bonds. Weak intermolecular forces between chains allow the polymer to be flexible.

MARK BREAKDOWN

Mark 1: Long carbon chain with covalent bonds

Mark 2: Repeating monomer units

Mark 3: Weak forces between chains

COMMON MISTAKES

- Ionic lattice structure
- No mention of intermolecular forces

EXAMINER TIP

Distinguish strong bonds within chains from weak forces between chains.

QUESTION 22

Explain why nanomaterials can have different properties to the bulk material.

QUESTION TYPE

Topic: Bonding, structure and properties | **Command word:** explain | **Marks:** 4 | **Skill tested:** Nanoparticles

WHAT THE QUESTION IS ASKING

Discuss high surface area to volume ratio.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Nanoparticles have a much higher surface area to volume ratio, so more atoms are on the surface and available to react, making them more reactive and giving different optical and catalytic properties.

MARK BREAKDOWN

Mark 1: Very small particles (1–100 nm)

Mark 2: High surface area to volume ratio

Mark 3: More surface atoms exposed

Mark 4: Different reactivity and properties result

COMMON MISTAKES

- Saying chemical bonds change type
- Ignoring surface area

EXAMINER TIP

Always link nanoparticle size to surface area ratio.

GRADE 7–9 EXTENSION

Give a specific example such as titanium dioxide in sunscreens.

QUESTION 23

Describe the structure and bonding in graphite and explain two uses based on its properties.

QUESTION TYPE

Topic: Bonding, structure and properties | **Command word:** describe | **Marks:** 4 | **Skill tested:** Giant covalent structures

WHAT THE QUESTION IS ASKING

Layers, weak forces between layers, delocalised electrons; link to lubricant and electrodes.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Graphite has hexagonal layers of carbon with weak forces between layers, so layers slide — useful as a lubricant. Delocalised electrons between layers allow it to conduct electricity — useful for electrodes.

MARK BREAKDOWN

Mark 1: Hexagonal layers of carbon

Mark 2: Weak intermolecular forces between layers → lubricant

Mark 3: Delocalised electrons → conducts electricity → electrodes

Mark 4: Strong covalent bonds within layers

COMMON MISTAKES

- Saying graphite is ionic
- Confusing with diamond uses

EXAMINER TIP

Structure → property → use for each point.

QUESTION 24

Compare ionic, simple covalent and giant covalent structures. In your answer, describe bonding, structure and how these affect melting point and electrical conductivity.

QUESTION TYPE

Topic: Bonding, structure and properties | **Command word:** compare | **Marks:** 6 | **Skill tested:** Structure and properties

WHAT THE QUESTION IS ASKING

Extended comparison of three structure types with properties.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Ionic: giant lattice of ions, strong electrostatic forces; high mp; conducts when molten or in solution. Simple covalent: small molecules, weak intermolecular forces; low mp; does not conduct. Giant covalent: atoms joined by strong covalent bonds in a network; very high mp; usually no conduct (except graphite).

MARK BREAKDOWN

Mark 1: Ionic: lattice, high mp, conducts molten/aqueous

Mark 2: Simple covalent: molecules, low mp, no conduct

Mark 3: Giant covalent: network, very high mp

Mark 4: Graphite exception for conductivity

Mark 5: Bonding types identified

Mark 6: Properties linked to structure

COMMON MISTAKES

- Mixing up intermolecular and covalent bonds
- Missing one structure type

EXAMINER TIP

Use a table mentally: structure, bonding, mp, conductivity.

GRADE 7–9 EXTENSION

Mention diamond vs graphite conductivity contrast.

QUESTION 25

Diamond and graphite are both forms of carbon. A sample of diamond has a mass of 12 g. Calculate the number of moles of carbon atoms. (Relative atomic mass of C = 12)

QUESTION TYPE

Topic: Bonding, structure and properties | **Command word:** calculate | **Marks:** 3 | **Skill tested:** Calculation

WHAT THE QUESTION IS ASKING

Use $\text{moles} = \text{mass} \div \text{Mr}$ for an element.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

$\text{Moles} = \text{mass} \div \text{Mr} = 12 \div 12 = 1.0 \text{ mol.}$

MARK BREAKDOWN

Mark 1: Recall $\text{moles} = \text{mass}/\text{Mr}$

Mark 2: $12 \div 12 = 1.0$

Mark 3: Unit: mol

COMMON MISTAKES

- Using 12×12
- Wrong Mr

EXAMINER TIP

For elements, Mr equals Ar.

QUESTION 26

A student investigates whether a unknown solid conducts electricity when solid and when dissolved in water. Describe what the results would show if the solid is an ionic compound.

QUESTION TYPE

Topic: Bonding, structure and properties | **Command word:** describe | **Marks:** 4 | **Skill tested:** Ionic compounds

WHAT THE QUESTION IS ASKING

Predict conductivity: poor solid, good aqueous.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Solid: no bulb lights — ions fixed in lattice. Aqueous solution: bulb lights — ions are free to move and carry charge.

MARK BREAKDOWN

- Mark 1: No conduction when solid — ions cannot move
- Mark 2: Conduction in solution — ions mobile
- Mark 3: Links result to ionic lattice
- Mark 4: Contrasts solid and aqueous states

COMMON MISTAKES

- Expecting conduction in solid
- Attributing conduction to electrons in solution

EXAMINER TIP

Always connect observation to particle mobility.

Section 2: Quantitative chemistry

QUESTION 27

State what is meant by the relative formula mass (M_r) of a compound.

QUESTION TYPE

Topic: Quantitative chemistry | Command word: state | Marks: 1 | Skill tested: Calculations

WHAT THE QUESTION IS ASKING

Define M_r as the sum of relative atomic masses.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The sum of the relative atomic masses of all the atoms shown in the formula.

MARK BREAKDOWN

Mark 1: The sum of the relative atomic masses of all the atoms shown in the formula.

COMMON MISTAKES

- Mass of one mole in grams only
- Average mass of isotopes

EXAMINER TIP

M_r has no units — it is a relative value.

QUESTION 28

Give the unit of concentration when it is measured in moles per cubic decimetre.

QUESTION TYPE

Topic: Quantitative chemistry | Command word: give | Marks: 1 | Skill tested: Concentration

WHAT THE QUESTION IS ASKING

Recall the standard concentration unit.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

mol/dm^3 (or mol dm^{-3}).

MARK BREAKDOWN

Mark 1: mol/dm^3 (or mol dm^{-3}).

COMMON MISTAKES

- g/dm^3 only
- mol/dm^2

EXAMINER TIP

Write the unit exactly as mol/dm^3 .

QUESTION 29

State the formula linking moles, mass and relative formula mass.

QUESTION TYPE

Topic: Quantitative chemistry | Command word: state | Marks: 1 | Skill tested: Moles

WHAT THE QUESTION IS ASKING

Recall the mole triangle formula.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

moles = mass $\div$ M_r

MARK BREAKDOWN

Mark 1: moles = mass $\div$ M_r

COMMON MISTAKES

- moles = mass $\times$ M_r
- moles = mass + M_r

EXAMINER TIP

Say 'divide mass by M_r'.

QUESTION 30

Describe how you would prepare 250 cm³ of a standard solution from a solid solute.

QUESTION TYPE

Topic: Quantitative chemistry | Command word: describe | Marks: 2 | Skill tested: Titration preparation

WHAT THE QUESTION IS ASKING

Outline weigh, dissolve, transfer, make up to volume.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Weigh the solid accurately, dissolve in a small volume of distilled water in a beaker, transfer to a volumetric flask, rinse beaker into flask, then add distilled water to the mark and invert to mix.

MARK BREAKDOWN

Mark 1: Weigh solute accurately on balance

Mark 2: Dissolve and make up to 250 cm³ in a volumetric flask

COMMON MISTAKES

- Using a measuring cylinder for final volume
- Adding solid directly to the flask without dissolving

EXAMINER TIP

Mention volumetric flask and making up to the mark.

QUESTION 31

Explain why the actual yield of a reaction is often less than the theoretical yield.

QUESTION TYPE

Topic: Quantitative chemistry | Command word: explain | Marks: 2 | Skill tested: Yield

WHAT THE QUESTION IS ASKING

Discuss incomplete reaction, side reactions, product lost.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The reaction may be incomplete, side reactions may form other products, and some product may be lost during separation or transfer.

MARK BREAKDOWN

Mark 1: Incomplete reaction

Mark 2: Side reactions or practical losses during transfer

COMMON MISTAKES

- Claiming atoms are destroyed
- Only giving one vague reason

EXAMINER TIP

Give at least two specific reasons.

QUESTION 32

Give the meaning of a balanced chemical equation in terms of particles.

QUESTION TYPE

Topic: Quantitative chemistry | Command word: give | Marks: 2 | Skill tested: Equations

WHAT THE QUESTION IS ASKING

State equal numbers of atoms and mole ratios.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Equal numbers of atoms of each element on both sides, showing the mole ratio in which reactants react and products form.

MARK BREAKDOWN

Mark 1: Equal numbers of atoms of each element on both sides

Mark 2: Shows mole ratio of reactants and products

COMMON MISTAKES

- Equal masses on both sides
- Only mentioning atoms without moles

EXAMINER TIP

Link balancing to conservation of mass.

QUESTION 33

A student titrates hydrochloric acid with sodium hydroxide solution. Explain why a pipette is used for the alkali and a burette for the acid.

QUESTION TYPE

Topic: Quantitative chemistry | Command word: explain | Marks: 3 | Skill tested: Titration

WHAT THE QUESTION IS ASKING

Link pipette to accurate fixed volume and burette to variable volume added.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A pipette delivers an accurate, fixed volume of alkali. A burette allows the acid volume to be added gradually and read accurately until the endpoint.

MARK BREAKDOWN

- Mark 1: Pipette: accurate fixed volume of alkali
- Mark 2: Burette: variable volume of acid measured
- Mark 3: Enables accurate titre reading

COMMON MISTAKES

- Saying both measure the same way
- Using a measuring cylinder instead

EXAMINER TIP

Contrast fixed volume (pipette) with variable volume (burette).

QUESTION 34

Describe how you would calculate the concentration of an acid from titration results.

QUESTION TYPE

Topic: Quantitative chemistry | Command word: describe | Marks: 3 | Skill tested: Titration calculations

WHAT THE QUESTION IS ASKING

Outline moles from known alkali, ratio, moles acid, concentration.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Find moles of alkali (concentration $\times$ volume). Use the equation ratio to find moles of acid. Divide moles of acid by acid volume in dm^3 to get concentration.

MARK BREAKDOWN

Mark 1: Moles alkali = $c \times V$

Mark 2: Use balanced equation ratio

Mark 3: Concentration acid = moles acid $\div$ volume acid

COMMON MISTAKES

- Skipping the mole ratio step
- Forgetting to convert cm^3 to dm^3

EXAMINER TIP

Always convert volumes to dm^3 before dividing.

QUESTION 35

Explain what is meant by atom economy and why a high atom economy is desirable in sustainable chemistry.

QUESTION TYPE

Topic: Quantitative chemistry | Command word: explain | Marks: 4 | Skill tested: Atom economy

WHAT THE QUESTION IS ASKING

Define atom economy and link to waste minimisation.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Atom economy is the percentage of reactant atoms that end up in the desired product. High atom economy means less waste, fewer by-products and more efficient use of resources.

MARK BREAKDOWN

Mark 1: Definition: useful product mass $\div$ total reactant mass $\times 100$

Mark 2: High atom economy $\rightarrow$ less waste

Mark 3: More sustainable / efficient

Mark 4: Fewer atoms wasted as by-products

COMMON MISTAKES

- Confusing with percentage yield
- No link to sustainability

EXAMINER TIP

Distinguish atom economy (theoretical) from percentage yield (actual).

GRADE 7–9 EXTENSION

Compare addition vs substitution reactions for atom economy.

QUESTION 36

A reaction has a theoretical yield of 20 g but an actual yield of 16 g. Describe how to calculate the percentage yield and state two reasons the yield might be below 100%.

QUESTION TYPE

Topic: Quantitative chemistry | Command word: describe | Marks: 4 | Skill tested: Percentage yield

WHAT THE QUESTION IS ASKING

Give formula and practical reasons for low yield.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Percentage yield = (actual yield ÷ theoretical yield) × 100 = (16 ÷ 20) × 100 = 80%. Reasons: incomplete reaction and product lost during filtration or transfer.

MARK BREAKDOWN

- Mark 1: Formula: actual/theoretical × 100
Mark 2: Calculation: 80%
Mark 3: Reason: incomplete reaction
Mark 4: Reason: practical loss

COMMON MISTAKES

- Swapping actual and theoretical
- Only one reason given

EXAMINER TIP

Show the calculation even when describing the method.

QUESTION 37

A factory produces ethanol by fermentation and by reacting ethene with steam. Compare the two methods in terms of raw materials, conditions, atom economy and sustainability.

QUESTION TYPE

Topic: Quantitative chemistry | Command word: compare | Marks: 6 | Skill tested: Industrial chemistry

WHAT THE QUESTION IS ASKING

Extended comparison of fermentation vs hydration of ethene.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Fermentation uses renewable sugar from crops, low temperature, batch process, moderate atom economy, slower. Hydration uses ethene from crude oil (finite), high temperature with catalyst, continuous process, high atom economy, faster but depends on fossil fuels.

MARK BREAKDOWN

- Mark 1: Fermentation: renewable, batch, lower rate

- Mark 2: Hydration: fossil fuel feedstock, continuous
- Mark 3: Atom economy comparison
- Mark 4: Sustainability: renewable vs finite resources
- Mark 5: Conditions compared
- Mark 6: Economic and environmental factors

COMMON MISTAKES

- Only describing one method
- Confusing atom economy with yield

EXAMINER TIP

Structure as two columns: fermentation vs hydration.

GRADE 7–9 EXTENSION

Discuss life-cycle analysis and carbon footprint.

QUESTION 38

Calculate the relative formula mass of calcium carbonate, CaCO_3 . (A_r: Ca = 40, C = 12, O = 16)

QUESTION TYPE

Topic: Quantitative chemistry | Command word: calculate | Marks: 3 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Add relative atomic masses for all atoms in the formula.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

$M_r = 40 + 12 + (3 \times 16) = 40 + 12 + 48 = 100$.

MARK BREAKDOWN

- Mark 1: Correct formula: $40 + 12 + 48$
- Mark 2: $M_r = 100$
- Mark 3: No units for M_r

COMMON MISTAKES

- Forgetting to multiply oxygen by 3
- Arithmetic error

EXAMINER TIP

Bracket the 3×16 before adding.

QUESTION 39

Describe how you would carry out a titration to find the concentration of hydrochloric acid using sodium hydroxide solution of known concentration.

QUESTION TYPE

Topic: Quantitative chemistry | Command word: describe | Marks: 4 | Skill tested: Required practical — titration

WHAT THE QUESTION IS ASKING

Full titration method with indicator and readings.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Rinse pipette with alkali and transfer 25.0 cm^3 to a conical flask. Add a few drops of indicator. Fill a burette with acid and record initial reading. Add acid slowly with swirling until the indicator changes colour. Record final reading and repeat for concordant titres.

MARK BREAKDOWN

- Mark 1: Pipette alkali accurately
- Mark 2: Indicator added to flask
- Mark 3: Burette for acid with initial/final readings
- Mark 4: Swirl and add dropwise near endpoint

COMMON MISTAKES

- Adding indicator to burette
- No mention of concordant results

EXAMINER TIP

Include equipment, indicator and concordant titres.

Section 2: Chemical changes

QUESTION 40

State the products of the electrolysis of molten lead bromide.

QUESTION TYPE

Topic: Chemical changes | Command word: state | Marks: 1 | Skill tested: Electrolysis

WHAT THE QUESTION IS ASKING

Recall products at each electrode for molten ionic compound.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Lead metal at the cathode and bromine gas at the anode.

MARK BREAKDOWN

Mark 1: Lead metal at the cathode and bromine gas at the anode.

COMMON MISTAKES

- Hydrogen at cathode
- Oxygen at anode

EXAMINER TIP

Molten — no water, so metal not hydrogen forms at cathode.

QUESTION 41

Give the name of the gas produced at the cathode when dilute sulfuric acid is electrolysed.

QUESTION TYPE

Topic: Chemical changes | Command word: give | Marks: 1 | Skill tested: Electrolysis of aqueous solutions

WHAT THE QUESTION IS ASKING

Identify cathode product with H^+ ions present.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Hydrogen.

MARK BREAKDOWN

Mark 1: Hydrogen.

COMMON MISTAKES

- Oxygen
- Sulfur dioxide

EXAMINER TIP

Less reactive than hydrogen at cathode? H^+ discharged.

QUESTION 42

State what happens to a metal atom when it forms a positive ion.

QUESTION TYPE

Topic: Chemical changes | Command word: state | Marks: 1 | Skill tested: Reactivity

WHAT THE QUESTION IS ASKING

Describe electron loss in oxidation.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

It loses one or more electrons.

MARK BREAKDOWN

Mark 1: It loses one or more electrons.

COMMON MISTAKES

- Gains electrons
- Shares electrons only

EXAMINER TIP

Oxidation is loss of electrons (OIL).

QUESTION 43

Describe the test for hydrogen gas.

QUESTION TYPE

Topic: Chemical changes | Command word: describe | Marks: 2 | Skill tested: Gas tests

WHAT THE QUESTION IS ASKING

Squeaky pop with lighted splint.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Hold a lighted splint at the mouth of the tube; hydrogen burns with a squeaky pop.

MARK BREAKDOWN

Mark 1: Lighted splint used

Mark 2: Squeaky pop observed

COMMON MISTAKES

- Using a glowing splint
- Limewater test

EXAMINER TIP

Only hydrogen gives the squeaky pop test.

QUESTION 44

Explain why aluminium is extracted by electrolysis but iron is extracted by reduction with carbon.

QUESTION TYPE

Topic: Chemical changes | **Command word:** explain | **Marks:** 2 | **Skill tested:** Extraction of metals

WHAT THE QUESTION IS ASKING

Link to reactivity series position.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Aluminium is more reactive than carbon, so carbon cannot displace it from its oxide; electrolysis is needed. Iron is less reactive than carbon, so carbon can reduce iron oxide.

MARK BREAKDOWN

- Mark 1: Al more reactive than C → electrolysis required
Mark 2: Fe less reactive than C → reduction with carbon works

COMMON MISTAKES

- Both extracted the same way
- Wrong reactivity order

EXAMINER TIP

Compare metal reactivity with carbon.

QUESTION 45

Give two observations when magnesium reacts with dilute hydrochloric acid.

QUESTION TYPE

Topic: Chemical changes | **Command word:** give | **Marks:** 2 | **Skill tested:** Acids and metals

WHAT THE QUESTION IS ASKING

Bubbles, heat, metal disappears.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Effervescence (bubbles of gas) and the magnesium ribbon disappears / flask feels warm.

MARK BREAKDOWN

- Mark 1: Bubbles of gas produced
Mark 2: Magnesium reacts/disappears or temperature rises

COMMON MISTAKES

- No gas produced

- Precipitate forms

EXAMINER TIP

Two distinct observations — gas and metal change.

QUESTION 46

Explain why the reactivity series can be used to predict whether a displacement reaction will occur.

QUESTION TYPE

Topic: Chemical changes | **Command word:** explain | **Marks:** 3 | **Skill tested:** Displacement reactions

WHAT THE QUESTION IS ASKING

More reactive metal displaces less reactive metal ion.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A more reactive metal will displace a less reactive metal from a solution of its salt because it loses electrons more readily and forms ions more easily.

MARK BREAKDOWN

- Mark 1: More reactive metal displaces less reactive
- Mark 2: Linked to ease of electron loss
- Mark 3: Reaction only occurs if metal is above other in series

COMMON MISTAKES

- Any metal displaces any salt
- Reactivity irrelevant

EXAMINER TIP

State the rule then apply to specific metals.

QUESTION 47

Describe what happens during the electrolysis of aqueous copper(II) sulfate using inert electrodes.

QUESTION TYPE

Topic: Chemical changes | **Command word:** describe | **Marks:** 3 | **Skill tested:** Electrolysis

WHAT THE QUESTION IS ASKING

Products and electrode changes with Cu^{2+} and H_2O present.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Copper is deposited at the cathode (pink-brown coating). Oxygen is produced at the anode. The blue solution fades as Cu^{2+} ions are removed.

MARK BREAKDOWN

Mark 1: Cu deposited at cathode

Mark 2: O_2 at anode

Mark 3: Solution colour fades

COMMON MISTAKES

- Hydrogen at cathode
- No change at electrodes

EXAMINER TIP

Link electrode products to ion discharge rules.

QUESTION 48

Explain the observations when iron nails are placed in copper(II) sulfate solution.

QUESTION TYPE

Topic: Chemical changes | **Command word:** explain | **Marks:** 4 | **Skill tested:** Displacement reactions

WHAT THE QUESTION IS ASKING

Iron displaces copper — colour change, coating, exothermic.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Iron is more reactive than copper, so it displaces copper from solution. The solution fades from blue, a brown coating of copper forms on the nail, and the temperature may rise.

MARK BREAKDOWN

Mark 1: Fe more reactive than Cu

Mark 2: Displacement occurs

Mark 3: Blue colour fades

Mark 4: Brown copper coating on nail

COMMON MISTAKES

- Copper displaces iron
- No visible change expected

EXAMINER TIP

Include colour changes and coating.

QUESTION 49

A student electrolyses copper(II) sulfate using copper electrodes. Describe what happens at each electrode and to the solution.

QUESTION TYPE

Topic: Chemical changes | Command word: describe | Marks: 4 | Skill tested: Electrolysis

WHAT THE QUESTION IS ASKING

Active electrodes — Cu dissolves at anode, deposits at cathode.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Anode: copper dissolves (mass decreases). Cathode: copper deposited (mass increases). Solution colour and concentration stay approximately constant as copper removed at anode equals copper deposited at cathode.

MARK BREAKDOWN

- Mark 1: Anode: Cu dissolves
- Mark 2: Cathode: Cu deposited
- Mark 3: Solution colour unchanged
- Mark 4: Mass transfer between electrodes

COMMON MISTAKES

- Oxygen at anode
- Solution turns colourless

EXAMINER TIP

Active copper electrodes behave differently from inert ones.

QUESTION 50

Describe how the reactivity series can be determined experimentally using metals and their salt solutions. Include at least three metals and explain how the results are interpreted.

QUESTION TYPE

Topic: Chemical changes | Command word: describe | Marks: 6 | Skill tested: Reactivity series

WHAT THE QUESTION IS ASKING

Plan displacement tests and build series from results.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Place clean metal pieces in solutions of other metal salts and observe whether displacement occurs. E.g. zinc in copper sulfate → brown coating (Zn more reactive). Copper in zinc sulfate → no reaction (Cu less reactive). Magnesium in copper sulfate → vigorous reaction (Mg most reactive of the three). No reaction means the metal is less reactive than the metal in the salt.

MARK BREAKDOWN

- Mark 1: Method: metal + salt solution
- Mark 2: At least three metals tested
- Mark 3: Observation: coating/colour change or no reaction
- Mark 4: Interpretation: reaction = more reactive
- Mark 5: Build ordered series from results
- Mark 6: Fair test: clean metals, same conditions

COMMON MISTAKES

- Only one metal tested
- Wrong interpretation of no reaction

EXAMINER TIP

Include both method and how to rank metals.

GRADE 7–9 EXTENSION

Discuss single displacement vs no reaction pairs.

QUESTION 51

Calculate the mass of magnesium needed to react completely with 100 cm³ of 0.5 mol/dm³ hydrochloric acid. $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$ (A_r: Mg = 24)

QUESTION TYPE

Topic: Chemical changes | Command word: calculate | Marks: 4 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Find moles HCl, use ratio, convert moles Mg to mass.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Moles HCl = $0.5 \times 0.1 = 0.05$ mol. Ratio 1:2 so moles Mg = 0.025 mol. Mass = $0.025 \times 24 = 0.6$ g.

MARK BREAKDOWN

- Mark 1: Moles HCl = 0.05
- Mark 2: Moles Mg = 0.025 from 1:2 ratio
- Mark 3: Mass = 0.025×24
- Mark 4: Final answer 0.6 g

COMMON MISTAKES

- Using 1:1 ratio
- Not converting cm³ to dm³

EXAMINER TIP

Check the balanced equation ratio before calculating.

QUESTION 52

Describe how you would electrolyse aqueous copper(II) sulfate using carbon electrodes and identify the products at each electrode.

QUESTION TYPE

Topic: Chemical changes | Command word: describe | Marks: 4 | Skill tested: Required practical — electrolysis

WHAT THE QUESTION IS ASKING

Set up cell, power supply, observe electrode changes.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Fill a beaker with copper(II) sulfate solution. Insert carbon electrodes connected to a low-voltage DC supply (anode +, cathode –). Observe pink-brown copper on cathode and bubbles at anode (oxygen). Test gases if required.

MARK BREAKDOWN

- Mark 1: DC power supply with correct electrode polarity
- Mark 2: Copper deposits on cathode
- Mark 3: Bubbles at anode
- Mark 4: Solution fades over time

COMMON MISTAKES

- Using AC supply
- Swapping electrode products

EXAMINER TIP

State electrode polarity and expected products.

Section 2: Energy changes

QUESTION 53

State whether bond breaking is endothermic or exothermic.

QUESTION TYPE

Topic: Energy changes | Command word: state | Marks: 1 | Skill tested: Energy changes

WHAT THE QUESTION IS ASKING

Recall energy absorbed to break bonds.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Endothermic (energy is taken in).

MARK BREAKDOWN

Mark 1: Endothermic (energy is taken in).

COMMON MISTAKES

- Exothermic
- Neither

EXAMINER TIP

Breaking bonds costs energy — endothermic.

QUESTION 54

Give the name for a reaction that releases energy to the surroundings.

QUESTION TYPE

Topic: Energy changes | Command word: give | Marks: 1 | Skill tested: Exothermic and endothermic

WHAT THE QUESTION IS ASKING

Define exothermic reaction.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Exothermic reaction.

MARK BREAKDOWN

Mark 1: Exothermic reaction.

COMMON MISTAKES

- Endothermic
- Neutralisation only

EXAMINER TIP

Exothermic = exits to surroundings.

QUESTION 55

State what a cell or battery converts chemical energy into.

QUESTION TYPE

Topic: Energy changes | Command word: state | Marks: 1 | Skill tested: Cells and batteries

WHAT THE QUESTION IS ASKING

Energy conversion in electrochemical cells.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Electrical energy.

MARK BREAKDOWN

Mark 1: Electrical energy.

COMMON MISTAKES

- Kinetic energy
- Nuclear energy

EXAMINER TIP

Cells convert chemical → electrical.

QUESTION 56

Describe the energy change when water freezes.

QUESTION TYPE

Topic: Energy changes | Command word: describe | Marks: 2 | Skill tested: Energy transfers

WHAT THE QUESTION IS ASKING

Exothermic — energy released as bonds form.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Freezing is exothermic: energy is transferred to the surroundings as intermolecular bonds form and temperature can stay constant at 0 °C.

MARK BREAKDOWN

Mark 1: Energy released to surroundings

Mark 2: Bonds forming is exothermic

COMMON MISTAKES

- Endothermic freezing
- No energy transfer

EXAMINER TIP

Bond forming releases energy.

QUESTION 57

Explain why a reaction feels cold to the touch.

QUESTION TYPE

Topic: Energy changes | Command word: explain | Marks: 2 | Skill tested: Endothermic reactions

WHAT THE QUESTION IS ASKING

Energy absorbed from surroundings including skin.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The reaction is endothermic, so it absorbs thermal energy from the surroundings (including your hand), making it feel cold.

MARK BREAKDOWN

Mark 1: Endothermic — absorbs energy from surroundings

Mark 2: Thermal energy removed from hand

COMMON MISTAKES

- Energy released
- Temperature always rises

EXAMINER TIP

Link cold feel to energy absorbed from surroundings.

QUESTION 58

Give two uses of exothermic reactions.

QUESTION TYPE

Topic: Energy changes | Command word: give | Marks: 2 | Skill tested: Applications

WHAT THE QUESTION IS ASKING

Hand warmers, self-heating cans, combustion heating.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Hand warmers and combustion in fuels for heating / transport.

MARK BREAKDOWN

Mark 1: Hand warmers or self-heating cans

Mark 2: Combustion for heating or engines

COMMON MISTAKES

- Endothermic examples
- Electrolysis

EXAMINER TIP

Both examples must be exothermic.

QUESTION 59

Describe how a simple calorimeter can be used to measure the temperature change of a reaction.

QUESTION TYPE

Topic: Energy changes | Command word: describe | Marks: 3 | Skill tested: Calorimetry

WHAT THE QUESTION IS ASKING

Polystyrene cup, measure T before and after, calculate ΔT .

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Measure a known volume of reactant temperature. Add second reactant in a polystyrene cup, stir gently, record maximum temperature change. Insulation reduces heat loss to surroundings.

MARK BREAKDOWN

- Mark 1: Measure initial temperature
- Mark 2: Mix in insulated cup and record max ΔT
- Mark 3: Polystyrene reduces heat loss

COMMON MISTAKES

- Metal beaker without insulation
- Only one temperature reading

EXAMINER TIP

Insulation and two temperature readings are essential.

QUESTION 60

Explain the difference between exothermic and endothermic reactions in terms of energy transfer.

QUESTION TYPE

Topic: Energy changes | Command word: explain | Marks: 3 | Skill tested: Energy changes

WHAT THE QUESTION IS ASKING

Direction of energy transfer between reaction and surroundings.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Exothermic: net energy released to surroundings, temperature often rises. Endothermic: net energy absorbed from surroundings, temperature often falls.

MARK BREAKDOWN

Mark 1: Exothermic releases energy to surroundings

Mark 2: Endothermic absorbs from surroundings

Mark 3: Link to temperature change

COMMON MISTAKES

- Both release energy
- Confusing with bond breaking only

EXAMINER TIP

State direction of energy transfer for each.

QUESTION 61

A student mixes anhydrous copper(II) sulfate with water. The flask becomes warm. Explain what type of reaction this is and what has happened to the energy.

QUESTION TYPE

Topic: Energy changes | Command word: explain | Marks: 4 | Skill tested: Energy changes

WHAT THE QUESTION IS ASKING

Exothermic hydration — energy released forming bonds.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The reaction is exothermic. Energy is released as new bonds form when water hydrates the anhydrous salt, transferring thermal energy to the surroundings and warming the flask.

MARK BREAKDOWN

Mark 1: Exothermic identified

Mark 2: Energy released as bonds form

Mark 3: Thermal energy transferred to surroundings

Mark 4: Temperature rise observed

COMMON MISTAKES

- Endothermic classification
- Energy destroyed

EXAMINER TIP

Connect warm flask to exothermic energy release.

QUESTION 62

Describe the reaction profile for an exothermic reaction, labelling reactants, products, activation energy and overall energy change.

QUESTION TYPE

Topic: Energy changes | **Command word:** describe | **Marks:** 4 | **Skill tested:** Reaction profiles

WHAT THE QUESTION IS ASKING

Draw/description: products lower than reactants, E_a hump.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Reactants start at higher energy. Activation energy is the peak before products form. Products are at lower energy than reactants. Overall energy change (ΔH) is negative (energy released).

MARK BREAKDOWN

- Mark 1: Reactants higher energy than products
- Mark 2: Activation energy shown as peak
- Mark 3: ΔH negative / energy released
- Mark 4: Correct labels on profile

COMMON MISTAKES

- Products higher than reactants
- No activation energy

EXAMINER TIP

Products lower = exothermic on a profile diagram.

QUESTION 63

Compare exothermic and endothermic reactions. In your answer, describe energy transfers, give examples, and explain how reaction profiles differ.

QUESTION TYPE

Topic: Energy changes | **Command word:** compare | **Marks:** 6 | **Skill tested:** Energy changes

WHAT THE QUESTION IS ASKING

Full comparison with examples and profiles.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Exothermic releases energy (combustion, neutralisation); products lower on profile; ΔH negative. Endothermic absorbs energy (thermal decomposition of limestone, photosynthesis); products higher; ΔH positive. Both require activation energy to start.

MARK BREAKDOWN

- Mark 1: Exothermic releases, endothermic absorbs

Mark 2: Examples of each

Mark 3: Profile: products lower vs higher

Mark 4: Activation energy in both

Mark 5: ΔH sign correct

Mark 6: Temperature effects on surroundings

COMMON MISTAKES

- No examples
- Profiles described incorrectly

EXAMINER TIP

Include examples, profiles and energy transfer.

GRADE 7–9 EXTENSION

Discuss bond breaking vs bond making energetics.

QUESTION 64

A fuel releases 840 kJ when 2.0 g burns completely. Calculate the energy released per gram. (2 marks)

QUESTION TYPE

Topic: Energy changes | Command word: calculate | Marks: 2 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Divide total energy by mass burned.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Energy per gram = $840 \div 2.0 = 420 \text{ kJ/g}$.

MARK BREAKDOWN

Mark 1: $840 \div 2.0$

Mark 2: 420 kJ/g

COMMON MISTAKES

- Multiplying instead of dividing
- Wrong unit

EXAMINER TIP

Include units in the final answer.

QUESTION 65

Describe how you would investigate whether dissolving ammonium nitrate in water is exothermic or endothermic.

QUESTION TYPE

Topic: Energy changes | **Command word:** describe | **Marks:** 4 | **Skill tested:** Required practical — temperature change

WHAT THE QUESTION IS ASKING

Measure temperature before and after dissolving in insulated cup.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Measure 50 cm³ water temperature in a polystyrene cup. Add a known mass of ammonium nitrate, stir until dissolved, record lowest temperature. A temperature drop indicates an endothermic process.

MARK BREAKDOWN

- Mark 1: Initial temperature recorded
- Mark 2: Dissolve in polystyrene cup
- Mark 3: Temperature drop → endothermic
- Mark 4: Repeat for reliability

COMMON MISTAKES

- No insulation
- Only one reading

EXAMINER TIP

Temperature fall = endothermic for dissolution.

Section 2: Rates of reaction

QUESTION 66

State one factor that increases the rate of a chemical reaction.

QUESTION TYPE

Topic: Rates of reaction | Command word: state | Marks: 1 | Skill tested: Factors affecting rate

WHAT THE QUESTION IS ASKING

Recall temperature, concentration, surface area, catalyst, pressure.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Increasing temperature (also accept: concentration, surface area, catalyst, pressure for gases).

MARK BREAKDOWN

Mark 1: Increasing temperature (also accept: concentration, surface area, catalyst, pressure for gases).

COMMON MISTAKES

- Decreasing temperature
- Adding inhibitor without context

EXAMINER TIP

Any one valid factor scores the mark.

QUESTION 67

Give the definition of a catalyst.

QUESTION TYPE

Topic: Rates of reaction | Command word: give | Marks: 1 | Skill tested: Catalysts

WHAT THE QUESTION IS ASKING

Define catalyst — speeds up, not used up.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A substance that increases the rate of a reaction without being used up.

MARK BREAKDOWN

Mark 1: A substance that increases the rate of a reaction without being used up.

COMMON MISTAKES

- Gets used up in reaction
- Slows the reaction

EXAMINER TIP

Include 'not used up' in your definition.

QUESTION 68

State what is meant by the term rate of reaction.

QUESTION TYPE

Topic: Rates of reaction | Command word: state | Marks: 1 | Skill tested: Rate of reaction

WHAT THE QUESTION IS ASKING

Define rate as change per unit time.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The speed at which reactants are used up or products are formed.

MARK BREAKDOWN

Mark 1: The speed at which reactants are used up or products are formed.

COMMON MISTAKES

- Always constant
- Only applies to gases

EXAMINER TIP

Mention reactants or products and speed.

QUESTION 69

Describe how increasing the surface area of a solid reactant affects the rate of reaction.

QUESTION TYPE

Topic: Rates of reaction | Command word: describe | Marks: 2 | Skill tested: Surface area

WHAT THE QUESTION IS ASKING

More area → more frequent collisions.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Breaking the solid into smaller pieces increases surface area, so more particles are exposed and collision frequency increases, increasing the rate.

MARK BREAKDOWN

Mark 1: Greater surface area

Mark 2: More frequent collisions → faster rate

COMMON MISTAKES

- Fewer collisions
- Surface area has no effect

EXAMINER TIP

Link surface area to collision frequency.

QUESTION 70

Explain why a reaction goes faster at a higher temperature.

QUESTION TYPE

Topic: Rates of reaction | Command word: explain | Marks: 2 | Skill tested: Temperature

WHAT THE QUESTION IS ASKING

Particles faster, more collisions, more exceed E_a .

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Particles have more kinetic energy, move faster, collide more frequently, and a greater proportion of collisions have energy above the activation energy.

MARK BREAKDOWN

Mark 1: Particles move faster / more kinetic energy

Mark 2: More collisions exceed activation energy

COMMON MISTAKES

- Only mentioning faster movement without activation energy
- Temperature decreases rate

EXAMINER TIP

Include both collision frequency and activation energy.

QUESTION 71

Give two ways to measure the rate of the reaction between magnesium and acid.

QUESTION TYPE

Topic: Rates of reaction | Command word: give | Marks: 2 | Skill tested: Measuring rates

WHAT THE QUESTION IS ASKING

Gas volume over time or mass loss over time.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Measure volume of hydrogen gas produced at regular time intervals, or measure mass of flask contents decreasing as gas escapes.

MARK BREAKDOWN

Mark 1: Gas volume collected over time

Mark 2: Mass loss of flask over time

COMMON MISTAKES

- Measure colour change only
- Single measurement with no time

EXAMINER TIP

Rate needs a quantity measured over time.

QUESTION 72

Explain how a catalyst increases the rate of reaction.

QUESTION TYPE

Topic: Rates of reaction | Command word: explain | Marks: 3 | Skill tested: Catalysts

WHAT THE QUESTION IS ASKING

Lower activation energy, more successful collisions.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A catalyst provides an alternative reaction pathway with lower activation energy, so a greater proportion of collisions are successful without being used up.

MARK BREAKDOWN

Mark 1: Alternative pathway

Mark 2: Lower activation energy

Mark 3: More successful collisions

COMMON MISTAKES

- Catalyst is used up
- Increases activation energy

EXAMINER TIP

Lower E_a is the key scientific explanation.

QUESTION 73

A student plots a graph of volume of gas against time for a reaction. Describe the shape of the curve and what it shows about the rate.

QUESTION TYPE

Topic: Rates of reaction | Command word: describe | Marks: 3 | Skill tested: Graphs

WHAT THE QUESTION IS ASKING

Steep at start, levels off — rate decreases as reactants used.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The curve rises steeply at first then levels off. Rate is fastest at the start when reactant concentration is highest, then slows as reactants are used up until reaction stops.

MARK BREAKDOWN

Mark 1: Steep initial gradient = fast rate

Mark 2: Curve levels off when reaction complete

Mark 3: Rate decreases over time

COMMON MISTAKES

- Constant gradient throughout
- Rate increases at end

EXAMINER TIP

Gradient = rate; levelling off = reaction finished.

QUESTION 74

Explain, using collision theory, why increasing the concentration of a reactant increases the rate.

QUESTION TYPE

Topic: Rates of reaction | Command word: explain | Marks: 4 | Skill tested: Collision theory

WHAT THE QUESTION IS ASKING

More particles per unit volume → more collisions.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Higher concentration means more particles per unit volume, so collisions between reacting particles occur more frequently, increasing the rate.

MARK BREAKDOWN

Mark 1: More particles per dm^3

Mark 2: More frequent collisions

Mark 3: Rate increases

Mark 4: Linked to collision theory

COMMON MISTAKES

- Particles move faster at same temperature
- Concentration reduces activation energy

EXAMINER TIP

Collision theory focuses on frequency at same temperature.

QUESTION 75

A student compares marble chips and powdered marble reacting with acid. Describe the expected results and explain them.

QUESTION TYPE

Topic: Rates of reaction | Command word: describe | Marks: 4 | Skill tested: Surface area practical

WHAT THE QUESTION IS ASKING

Powder faster — greater surface area.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Powdered marble reacts faster, producing gas more quickly (steeper graph). Powder has greater surface area, so more acid contacts calcium carbonate per second, increasing collision frequency.

MARK BREAKDOWN

- Mark 1: Powder reacts faster
- Mark 2: Greater surface area of powder
- Mark 3: More frequent collisions
- Mark 4: Steeper gas volume graph

COMMON MISTAKES

- Lumps faster than powder
- No link to collisions

EXAMINER TIP

Compare observations then explain with surface area.

QUESTION 76

Plan an investigation to find how temperature affects the rate of reaction between sodium thiosulfate and hydrochloric acid. Include equipment, method, safety, and how you would process results.

QUESTION TYPE

Topic: Rates of reaction | Command word: describe | Marks: 6 | Skill tested: Investigating rates

WHAT THE QUESTION IS ASKING

Cross formation timing at different temperatures.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Warm solutions to different temperatures (water baths). Mix fixed volumes of sodium thiosulfate and acid in a flask over a marked cross. Time until cross disappears. Repeat at each temperature. Plot time vs temperature; shorter time = faster rate. Wear goggles; acids are irritants.

MARK BREAKDOWN

Mark 1: Range of temperatures using water bath

Mark 2: Same volumes and concentrations

Mark 3: Time until cross obscured

Mark 4: Shorter time = faster rate

Mark 5: Repeat and calculate mean

Mark 6: Safety: goggles, acid handling

COMMON MISTAKES

- No control of other variables
- No graph or processing

EXAMINER TIP

Fair test + measurement + how to interpret results.

GRADE 7–9 EXTENSION

Discuss inverse relationship between time and rate.

QUESTION 77

In a rate experiment, 48 cm³ of gas is collected in 2 minutes. Calculate the mean rate of gas production in cm³/s.

QUESTION TYPE

Topic: Rates of reaction | Command word: calculate | Marks: 3 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Rate = volume ÷ time; convert minutes to seconds.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Time = 2 × 60 = 120 s. Rate = 48 ÷ 120 = 0.4 cm³/s.

MARK BREAKDOWN

Mark 1: Convert 2 min to 120 s

Mark 2: 48 ÷ 120 = 0.4

Mark 3: Unit cm³/s

COMMON MISTAKES

- Dividing by 2 instead of 120
- Wrong unit

EXAMINER TIP

Always convert time to seconds for cm³/s.

QUESTION 78

Describe how you would investigate the effect of concentration on the rate of reaction between magnesium ribbon and hydrochloric acid.

QUESTION TYPE

Topic: Rates of reaction | **Command word:** describe | **Marks:** 4 | **Skill tested:** Required practical — rates

WHAT THE QUESTION IS ASKING

Different acid concentrations, measure gas volume over time.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Use equal lengths of magnesium and different concentrations of acid at the same temperature. Collect hydrogen in a gas syringe at regular intervals or measure time for a fixed volume. Plot graphs and compare initial gradients.

MARK BREAKDOWN

- Mark 1: Independent variable: acid concentration
- Mark 2: Control: Mg size, temperature, volume
- Mark 3: Measure gas volume vs time
- Mark 4: Compare initial gradients

COMMON MISTAKES

- Changing temperature simultaneously
- No repeated measurements

EXAMINER TIP

State IV, CVs and how rate is measured.

Section 2: Organic chemistry

QUESTION 79

State the general formula of the alkane homologous series.

QUESTION TYPE

Topic: Organic chemistry | Command word: state | Marks: 1 | Skill tested: Alkanes

WHAT THE QUESTION IS ASKING

Recall C_nH_{2n+2} .

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

C_nH_{2n+2}

MARK BREAKDOWN

Mark 1: C_nH_{2n+2}

COMMON MISTAKES

- C_nH_n
- C_nH_{2n}

EXAMINER TIP

Alkanes are saturated: C_nH_{2n+2} .

QUESTION 80

Give the name of the functional group in alcohols.

QUESTION TYPE

Topic: Organic chemistry | Command word: give | Marks: 1 | Skill tested: Functional groups

WHAT THE QUESTION IS ASKING

Identify $-OH$ group.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Hydroxyl ($-OH$).

MARK BREAKDOWN

Mark 1: Hydroxyl ($-OH$).

COMMON MISTAKES

- Carboxyl
- Carbonyl only

EXAMINER TIP

Say hydroxyl or $-OH$ group.

QUESTION 81

State what is meant by a hydrocarbon.

QUESTION TYPE

Topic: Organic chemistry | Command word: state | Marks: 1 | Skill tested: Organic basics

WHAT THE QUESTION IS ASKING

Define compounds of C and H only.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A compound containing only hydrogen and carbon atoms.

MARK BREAKDOWN

Mark 1: A compound containing only hydrogen and carbon atoms.

COMMON MISTAKES

- Contains oxygen as well
- Only carbon

EXAMINER TIP

Only C and H — no other elements.

QUESTION 82

Describe the test for unsaturation using bromine water.

QUESTION TYPE

Topic: Organic chemistry | Command word: describe | Marks: 2 | Skill tested: Alkenes

WHAT THE QUESTION IS ASKING

Orange to colourless with alkenes.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Add bromine water to the hydrocarbon. With an alkene, the orange solution decolourises. With an alkane, the orange colour remains.

MARK BREAKDOWN

Mark 1: Bromine water turns from orange to colourless with alkenes

Mark 2: No change with alkanes

COMMON MISTAKES

- Limewater test
- Alkanes decolourise bromine

EXAMINER TIP

Alkenes decolourise; alkanes do not.

QUESTION 83

Explain why alkanes are described as saturated.

QUESTION TYPE

Topic: Organic chemistry | Command word: explain | Marks: 2 | Skill tested: Alkanes

WHAT THE QUESTION IS ASKING

Single bonds only — no more atoms can add.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

All carbon–carbon bonds are single bonds, so no more atoms can be added to the molecule.

MARK BREAKDOWN

Mark 1: Only single C–C bonds

Mark 2: Cannot take part in addition reactions easily

COMMON MISTAKES

- Contain water
- Have a functional group

EXAMINER TIP

Saturated = only single bonds between carbons.

QUESTION 84

Give the products of complete combustion of a hydrocarbon.

QUESTION TYPE

Topic: Organic chemistry | Command word: give | Marks: 2 | Skill tested: Combustion

WHAT THE QUESTION IS ASKING

CO₂ and H₂O only in complete combustion.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Carbon dioxide and water.

MARK BREAKDOWN

Mark 1: Carbon dioxide

Mark 2: Water

COMMON MISTAKES

- Carbon monoxide and water

- Carbon and hydrogen

EXAMINER TIP

Complete combustion needs plenty of oxygen.

QUESTION 85

Describe how ethanol can be produced by fermentation.

QUESTION TYPE

Topic: Organic chemistry | Command word: describe | Marks: 3 | Skill tested: Fermentation

WHAT THE QUESTION IS ASKING

Sugar, yeast, anaerobic, warm conditions.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Dissolve sugar in water, add yeast, and keep warm (about 30–40 °C) in the absence of air. Yeast enzymes convert glucose to ethanol and carbon dioxide.

MARK BREAKDOWN

Mark 1: Sugar/glucose substrate with yeast

Mark 2: Anaerobic warm conditions

Mark 3: Products ethanol and CO₂

COMMON MISTAKES

- Requires oxygen
- Uses ethene and steam

EXAMINER TIP

Anaerobic and yeast are essential keywords.

QUESTION 86

Explain why polymers are difficult to dispose of.

QUESTION TYPE

Topic: Organic chemistry | Command word: explain | Marks: 3 | Skill tested: Polymers

WHAT THE QUESTION IS ASKING

Non-biodegradable, persist in landfill, pollution.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Most polymers are non-biodegradable because bacteria cannot break the strong covalent bonds in the chains, so they persist in landfill or as litter, causing pollution.

MARK BREAKDOWN

Mark 1: Strong C–C bonds resist breakdown

Mark 2: Non-biodegradable

Mark 3: Environmental persistence / pollution

COMMON MISTAKES

- Biodegrade quickly
- Dissolve in water easily

EXAMINER TIP

Link strong bonds to lack of biodegradability.

QUESTION 87

Describe the addition reaction of ethene with bromine and draw the structural change.

QUESTION TYPE

Topic: Organic chemistry | Command word: describe | Marks: 4 | Skill tested: Alkenes

WHAT THE QUESTION IS ASKING

C=C breaks, Br₂ adds across bond.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The C=C double bond opens and bromine atoms add across it, forming dibromoethane: CH₂=CH₂ + Br₂ → CH₂BrCH₂Br. Orange bromine decolourises.

MARK BREAKDOWN

- Mark 1: Double bond opens
- Mark 2: Br adds across carbons
- Mark 3: Product dibromoethane
- Mark 4: Decolourisation observed

COMMON MISTAKES

- Hydrogen halide substitution
- Alkane product

EXAMINER TIP

Addition not substitution — double bond breaks.

QUESTION 88

Compare the production of ethanol by fermentation and by reacting ethene with steam.

QUESTION TYPE

Topic: Organic chemistry | Command word: compare | Marks: 4 | Skill tested: Ethanol production

WHAT THE QUESTION IS ASKING

Feedstock, conditions, batch vs continuous.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.

4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Fermentation uses renewable glucose, low temperature, batch process, slower, dilute ethanol produced.
Hydration uses ethene from crude oil, high temperature with acid catalyst, continuous, faster, pure ethanol.

MARK BREAKDOWN

- Mark 1: Fermentation: renewable, batch, dilute
- Mark 2: Hydration: ethene, high T, catalyst
- Mark 3: Continuous vs batch
- Mark 4: Sustainability difference

COMMON MISTAKES

- Only one method described
- Wrong catalyst for hydration

EXAMINER TIP

Compare at least three features.

QUESTION 89

Crude oil is separated and processed to make useful fuels and polymers. Describe fractional distillation of crude oil and explain how cracking is used to meet demand for petrol and ethene.

QUESTION TYPE

Topic: Organic chemistry | Command word: describe | Marks: 6 | Skill tested: Crude oil

WHAT THE QUESTION IS ASKING

Fractionating column + cracking to smaller molecules.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Crude oil is heated and vapours enter a fractionating column. Fractions condense at different heights according to boiling point — gases at top, bitumen at bottom. Cracking breaks long alkane molecules into shorter alkanes and alkenes using heat and a catalyst, producing more petrol and ethene for polymers.

MARK BREAKDOWN

- Mark 1: Fractional distillation by boiling point
- Mark 2: Named fractions
- Mark 3: Cracking breaks long chains
- Mark 4: Produces shorter alkanes + alkenes
- Mark 5: Meets demand for fuels and feedstock
- Mark 6: Catalyst or steam used in cracking

COMMON MISTAKES

- Crude oil is a single compound
- Cracking joins small molecules

EXAMINER TIP

Cover both separation and cracking with purposes.

GRADE 7–9 EXTENSION

Discuss finite nature of crude oil.

QUESTION 90

The displayed formula of butane is C_4H_{10} . Calculate its relative formula mass. (A_r: C = 12, H = 1)

QUESTION TYPE

Topic: Organic chemistry | Command word: calculate | Marks: 2 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Sum A_r values for all atoms in formula.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

$M_r = (4 \times 12) + (10 \times 1) = 48 + 10 = 58$.

MARK BREAKDOWN

Mark 1: $4 \times 12 + 10 \times 1$

Mark 2: $M_r = 58$

COMMON MISTAKES

- Using $4 + 10$ only
- Wrong multiplication

EXAMINER TIP

Count every atom in the displayed formula.

QUESTION 91

Describe how you would test a liquid to show it contains an unsaturated hydrocarbon.

QUESTION TYPE

Topic: Organic chemistry | Command word: describe | Marks: 4 | Skill tested: Test for alkenes

WHAT THE QUESTION IS ASKING

Bromine water decolourisation test.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Add a few drops of orange bromine water to the sample in a test tube and shake. If the orange colour disappears, an unsaturated hydrocarbon (alkene) is present.

MARK BREAKDOWN

Mark 1: Add bromine water

Mark 2: Shake mixture

Mark 3: Decolourisation = unsaturated

Mark 4: Control comparison if possible

COMMON MISTAKES

- Use limewater
- Expect colour to stay orange for alkene

EXAMINER TIP

State reagent, observation and conclusion.

Section 2: Chemical analysis

QUESTION 92

State the colour of copper(II) ions in solution.

QUESTION TYPE

Topic: Chemical analysis | Command word: state | Marks: 1 | Skill tested: Flame and gas tests

WHAT THE QUESTION IS ASKING

Recall blue solution for Cu^{2+} .

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Blue.

MARK BREAKDOWN

Mark 1: Blue.

COMMON MISTAKES

- Green
- Colourless

EXAMINER TIP

Cu^{2+} aqueous = blue.

QUESTION 93

Give the flame colour for potassium compounds.

QUESTION TYPE

Topic: Chemical analysis | Command word: give | Marks: 1 | Skill tested: Flame tests

WHAT THE QUESTION IS ASKING

Lilac flame for potassium.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Lilac.

MARK BREAKDOWN

Mark 1: Lilac.

COMMON MISTAKES

- Crimson
- Yellow

EXAMINER TIP

Potassium = lilac; do not confuse with sodium yellow.

QUESTION 94

State the test for carbon dioxide gas.

QUESTION TYPE

Topic: Chemical analysis | **Command word:** state | **Marks:** 1 | **Skill tested:** Gas tests

WHAT THE QUESTION IS ASKING

Limewater turns cloudy.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Bubble through limewater; it turns cloudy (milky).

MARK BREAKDOWN

Mark 1: Bubble through limewater; it turns cloudy (milky).

COMMON MISTAKES

- Limewater used
- Turns cloudy/milky

EXAMINER TIP

CO₂ + limewater is a standard test.

QUESTION 95

Describe the flame test for sodium ions.

QUESTION TYPE

Topic: Chemical analysis | **Command word:** describe | **Marks:** 2 | **Skill tested:** Flame tests

WHAT THE QUESTION IS ASKING

Clean loop, sample, yellow-orange flame.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Clean a nichrome or platinum wire loop in acid and flame, dip into the sample, and hold in a non-luminous Bunsen flame. Sodium produces a yellow-orange flame.

MARK BREAKDOWN

Mark 1: Clean wire loop to avoid contamination
Mark 2: Yellow-orange flame colour for sodium

COMMON MISTAKES

- Using a luminous flame
- Crimson colour quoted

EXAMINER TIP

Cleaning the loop prevents cross-contamination.

QUESTION 96

Explain why instrumental methods of analysis can be more sensitive than chemical tests.

QUESTION TYPE

Topic: Chemical analysis | Command word: explain | Marks: 2 | Skill tested: Instrumental analysis

WHAT THE QUESTION IS ASKING

Detect tiny amounts, fast, accurate.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Instruments such as mass spectrometers can detect very small quantities, give rapid results, and are highly accurate, whereas chemical tests may miss trace amounts.

MARK BREAKDOWN

Mark 1: Detect very small amounts

Mark 2: Fast and accurate compared to manual tests

COMMON MISTAKES

- Always cheaper
- Never need calibration

EXAMINER TIP

Sensitivity and accuracy are key advantages.

QUESTION 97

Give two ions identified by precipitation reactions with sodium hydroxide solution.

QUESTION TYPE

Topic: Chemical analysis | Command word: give | Marks: 2 | Skill tested: Metal hydroxide precipitates

WHAT THE QUESTION IS ASKING

Cu^{2+} blue, Fe^{2+} green, Fe^{3+} brown, Al^{3+} white then dissolves in excess.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Copper(II) ions (blue precipitate) and iron(III) ions (brown precipitate).

MARK BREAKDOWN

Mark 1: Cu^{2+} → blue precipitate

Mark 2: Fe^{3+} → brown precipitate

COMMON MISTAKES

- Sodium ions identified
- No precipitate mentioned

EXAMINER TIP

Name ion and precipitate colour.

QUESTION 98

Describe how paper chromatography can be used to separate and identify the dyes in a food colouring.

QUESTION TYPE

Topic: Chemical analysis | **Command word:** describe | **Marks:** 3 | **Skill tested:** Chromatography

WHAT THE QUESTION IS ASKING

Pencil line, solvent, R_f values.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Spot the colouring on pencil baseline on chromatography paper, place in solvent below the line, allow solvent to rise. Different dyes travel different distances; compare R_f values with known standards.

MARK BREAKDOWN

Mark 1: Baseline drawn in pencil

Mark 2: Solvent below spot level

Mark 3: Dyes separated and identified by R_f

COMMON MISTAKES

- Baseline in ink
- Solvent above spots

EXAMINER TIP

Pencil baseline and solvent level are critical.

QUESTION 99

A white precipitate forms when sodium hydroxide is added to a solution but dissolves in excess NaOH. Identify the metal ion and explain the observations.

QUESTION TYPE

Topic: Chemical analysis | **Command word:** explain | **Marks:** 3 | **Skill tested:** Metal hydroxide tests

WHAT THE QUESTION IS ASKING

Al³⁺ or Zn²⁺ — amphoteric hydroxides.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Aluminium ions (or zinc ions): a white precipitate of metal hydroxide forms, then dissolves in excess sodium hydroxide to form a soluble complex ion.

MARK BREAKDOWN

Mark 1: White precipitate with NaOH

Mark 2: Dissolves in excess NaOH

Mark 3: Ion identified as Al^{3+} or Zn^{2+} .

COMMON MISTAKES

- Copper ions
- Precipitate insoluble in excess

EXAMINER TIP

White ppt + dissolves in excess = Al^{3+} or Zn^{2+} .

QUESTION 100

Explain how a mass spectrometer can be used to determine relative atomic mass.

QUESTION TYPE

Topic: Chemical analysis | **Command word:** explain | **Marks:** 4 | **Skill tested:** Mass spectrometry

WHAT THE QUESTION IS ASKING

Ionise, accelerate, deflect by mass/charge.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Atoms are ionised, accelerated, and deflected by a magnetic field according to mass/charge ratio. The detector records peaks; the furthest major peak for an element gives the relative atomic mass.

MARK BREAKDOWN

Mark 1: Sample ionised

Mark 2: Deflection by m/z ratio

Mark 3: Detector produces spectrum

Mark 4: Link from main peak

COMMON MISTAKES

- Measures melting point
- Only works for compounds

EXAMINER TIP

Link peak position to relative atomic mass.

QUESTION 101

Describe tests to identify sulfate ions and chloride ions in separate solutions.

QUESTION TYPE

Topic: Chemical analysis | **Command word:** describe | **Marks:** 4 | **Skill tested:** Anion tests

WHAT THE QUESTION IS ASKING

BaCl₂ for sulfate; AgNO₃ for chloride.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Sulfate: add dilute hydrochloric acid then barium chloride solution — white precipitate of barium sulfate.

Chloride: add dilute nitric acid then silver nitrate — white precipitate of silver chloride.

MARK BREAKDOWN

- Mark 1: Sulfate: $\text{HCl} + \text{BaCl}_2 \rightarrow \text{white BaSO}_4 \text{ precipitate}$
- Mark 2: Chloride: $\text{HNO}_3 + \text{AgNO}_3 \rightarrow \text{white AgCl precipitate}$
- Mark 3: Acid added first to remove interfering ions
- Mark 4: Both tests described for separate solutions

COMMON MISTAKES

- No acid added first
- Wrong precipitate colours

EXAMINER TIP

Acid first prevents false positives.

QUESTION 102

A forensic scientist receives a sample of white powder. Describe a logical sequence of tests to identify whether it contains carbonate ions, chloride ions or starch.

QUESTION TYPE

Topic: Chemical analysis | **Command word:** describe | **Marks:** 6 | **Skill tested:** Chemical analysis

WHAT THE QUESTION IS ASKING

Plan tests in sensible order with expected results.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Add dilute acid — effervescence and CO₂ turning limewater cloudy indicates carbonate. To a fresh sample, add nitric acid then silver nitrate — white precipitate indicates chloride. To another sample, add iodine solution — blue-black colour indicates starch.

MARK BREAKDOWN

- Mark 1: Acid + limewater for carbonate/CO₂
- Mark 2: AgNO₃ test for chloride

Mark 3: Iodine test for starch

Mark 4: Logical order with controls

Mark 5: Expected positive results stated

Mark 6: Safety where relevant

COMMON MISTAKES

- One test for all ions
- Wrong reagents

EXAMINER TIP

Separate portions — one test per ion.

GRADE 7–9 EXTENSION

Discuss why acid is added before sulfate/chloride tests.

QUESTION 103

In chromatography, a dye travels 6.0 cm and the solvent front travels 10.0 cm. Calculate the R_f value.

QUESTION TYPE

Topic: Chemical analysis | Command word: calculate | Marks: 2 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

R_f = distance by dye $\div$ distance by solvent.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

$R_f = 6.0 \div 10.0 = 0.60$ (no units).

MARK BREAKDOWN

Mark 1: R_f = distance dye $\div$ distance solvent

Mark 2: 0.60

COMMON MISTAKES

- Dividing the wrong way
- Adding units

EXAMINER TIP

R_f has no units and must be ≤ 1 .

QUESTION 104

Describe how you would carry out paper chromatography to compare two ink samples.

QUESTION TYPE

Topic: Chemical analysis | Command word: describe | Marks: 4 | Skill tested: Required practical — chromatography

WHAT THE QUESTION IS ASKING

Baseline, solvent, compare R_f values.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.

2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Draw a pencil baseline on chromatography paper, spot each ink sample, place in a beaker with shallow solvent below the line, cover and allow to run. Mark solvent front, measure distances and calculate R_f values to compare.

MARK BREAKDOWN

- Mark 1: Pencil baseline
- Mark 2: Solvent below spots
- Mark 3: Measure distances
- Mark 4: Calculate and compare R_f

COMMON MISTAKES

- Ink baseline
- Solvent above spots

EXAMINER TIP

Never use ink for the baseline.

Section 2: Chemistry of the atmosphere

QUESTION 105

State the approximate percentage of nitrogen in the Earth's atmosphere.

QUESTION TYPE

Topic: Chemistry of the atmosphere | Command word: state | Marks: 1 | Skill tested: Earth's atmosphere

WHAT THE QUESTION IS ASKING

Recall ~78% nitrogen.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

About 78% (accept 78–79%).

MARK BREAKDOWN

Mark 1: About 78% (accept 78–79%).

COMMON MISTAKES

- 21%
- 1%

EXAMINER TIP

Nitrogen is the most abundant gas.

QUESTION 106

Give the main greenhouse gas produced by burning fossil fuels.

QUESTION TYPE

Topic: Chemistry of the atmosphere | Command word: give | Marks: 1 | Skill tested: Greenhouse gases

WHAT THE QUESTION IS ASKING

CO₂ from combustion.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Carbon dioxide.

MARK BREAKDOWN

Mark 1: Carbon dioxide.

COMMON MISTAKES

- Methane only
- Nitrogen

EXAMINER TIP

Fossil fuel combustion mainly produces CO₂.

QUESTION 107

State one product of complete combustion of methane.

QUESTION TYPE

Topic: Chemistry of the atmosphere | Command word: state | Marks: 1 | Skill tested: Combustion

WHAT THE QUESTION IS ASKING

CO₂ or H₂O from CH₄ + O₂.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Carbon dioxide (or water).

MARK BREAKDOWN

Mark 1: Carbon dioxide (or water).

COMMON MISTAKES

- Carbon monoxide
- Sulfur dioxide

EXAMINER TIP

Complete combustion of methane: CO₂ and H₂O.

QUESTION 108

Describe how the early Earth's atmosphere differed from today's atmosphere.

QUESTION TYPE

Topic: Chemistry of the atmosphere | Command word: describe | Marks: 2 | Skill tested: Evolution of atmosphere

WHAT THE QUESTION IS ASKING

High CO₂, little O₂, volcanic gases.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The early atmosphere had high carbon dioxide, little or no oxygen, and water vapour from volcanoes, whereas today it is mostly nitrogen and oxygen with little CO₂.

MARK BREAKDOWN

Mark 1: Early: high CO₂, little O₂

Mark 2: Today: mostly N₂ and O₂, low CO₂

COMMON MISTAKES

- Early atmosphere was mostly oxygen
- No volcanic gases mentioned

EXAMINER TIP

Contrast early vs modern composition.

QUESTION 109

Explain how an increase in greenhouse gases can cause global climate change.

QUESTION TYPE

Topic: Chemistry of the atmosphere | **Command word:** explain | **Marks:** 2 | **Skill tested:** Climate change

WHAT THE QUESTION IS ASKING

Trap IR radiation, rise in global temperature.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Greenhouse gases absorb infrared radiation emitted from Earth and re-emit it, reducing heat loss to space and causing average global temperatures to rise.

MARK BREAKDOWN

Mark 1: Absorb infrared radiation

Mark 2: Reduce heat loss to space / trap heat

COMMON MISTAKES

- Block sunlight entering
- Ozone depletion only

EXAMINER TIP

Focus on infrared trapping mechanism.

QUESTION 110

Give two human activities that increase atmospheric carbon dioxide.

QUESTION TYPE

Topic: Chemistry of the atmosphere | **Command word:** give | **Marks:** 2 | **Skill tested:** Carbon footprint

WHAT THE QUESTION IS ASKING

Burning fossil fuels and deforestation.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Burning fossil fuels for transport and power, and deforestation (reducing photosynthesis).

MARK BREAKDOWN

Mark 1: Burning fossil fuels

Mark 2: Deforestation

COMMON MISTAKES

- Recycling metals
- Using catalysts

EXAMINER TIP

Both must increase CO₂.

QUESTION 111

Describe how carbon dioxide was removed from the early atmosphere.

QUESTION TYPE

Topic: Chemistry of the atmosphere | **Command word:** describe | **Marks:** 3 | **Skill tested:** Carbon cycle history

WHAT THE QUESTION IS ASKING

Dissolved in oceans, photosynthesis, sedimentary rocks.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Carbon dioxide dissolved in oceans, was used in photosynthesis by early plants and algae forming organic matter, and formed carbonate precipitates that became sedimentary rocks.

MARK BREAKDOWN

Mark 1: Dissolved in oceans

Mark 2: Photosynthesis

Mark 3: Formation of sedimentary rocks/carbonates

COMMON MISTAKES

- Only combustion mentioned
- Oxygen removal

EXAMINER TIP

Three removal routes are commonly credited.

QUESTION 112

Explain why carbon monoxide is dangerous to humans.

QUESTION TYPE

Topic: Chemistry of the atmosphere | **Command word:** explain | **Marks:** 3 | **Skill tested:** Air pollutants

WHAT THE QUESTION IS ASKING

Binds to haemoglobin instead of oxygen.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Carbon monoxide binds strongly to haemoglobin in red blood cells, preventing oxygen transport, which can cause unconsciousness or death.

MARK BREAKDOWN

Mark 1: Binds to haemoglobin

Mark 2: Reduces oxygen transport

Mark 3: Toxic effect on body

COMMON MISTAKES

- Acid rain formation
- Greenhouse effect only

EXAMINER TIP

Link to haemoglobin, not just 'toxic'.

QUESTION 113

Describe how sulfur dioxide and oxides of nitrogen contribute to acid rain and suggest ways to reduce emissions.

QUESTION TYPE

Topic: Chemistry of the atmosphere | Command word: describe | Marks: 4 | Skill tested: Acid rain

WHAT THE QUESTION IS ASKING

Dissolve in rain forming acids; flue gas desulfurisation, catalytic converters.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

SO₂ and NO_x dissolve in atmospheric moisture forming sulfuric and nitric acids, lowering rain pH. Reduce SO₂ by flue gas desulfurisation; reduce NO_x with catalytic converters in vehicles and low-sulfur fuels.

MARK BREAKDOWN

Mark 1: SO₂/NO_x form acids in rain

Mark 2: Acid rain damages buildings and ecosystems

Mark 3: FGD for sulfur

Mark 4: Catalytic converters for vehicles

COMMON MISTAKES

- CO₂ causes acid rain directly
- No reduction methods

EXAMINER TIP

Cause + effect + two reduction strategies.

QUESTION 114

A car burns petrol in a poorly ventilated garage. Explain the gases that may build up and their hazards.

QUESTION TYPE

Topic: Chemistry of the atmosphere | **Command word:** explain | **Marks:** 4 | **Skill tested:** Incomplete combustion

WHAT THE QUESTION IS ASKING

CO toxic; CO replaces oxygen.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Incomplete combustion produces carbon monoxide, which is toxic as it binds to haemoglobin. Carbon dioxide also builds up, displacing oxygen and causing breathing difficulties.

MARK BREAKDOWN

- Mark 1: CO from incomplete combustion
- Mark 2: CO binds haemoglobin — toxic
- Mark 3: CO builds up displacing O₂
- Mark 4: Breathing difficulties from low oxygen

COMMON MISTAKES

- Only CO mentioned
- O₂ increases in garage

EXAMINER TIP

Distinguish CO toxicity from CO suffocation risk.

QUESTION 115

Evaluate the evidence that human activity is causing climate change. In your answer, describe greenhouse gases, their sources, and possible consequences and responses.

QUESTION TYPE

Topic: Chemistry of the atmosphere | **Command word:** evaluate | **Marks:** 6 | **Skill tested:** Climate change

WHAT THE QUESTION IS ASKING

Extended evaluation of anthropogenic climate change.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Evidence: rising CO₂ from fossil fuels and deforestation correlates with rising global temperatures. Greenhouse gases trap infrared radiation. Consequences: melting ice, sea level rise, extreme weather. Responses: renewable energy, carbon capture, international agreements, individual reduction of carbon footprint.

MARK BREAKDOWN

- Mark 1: Greenhouse effect explained

Mark 2: Human sources of CO₂ identified

Mark 3: Evidence of temperature rise

Mark 4: Consequences described

Mark 5: Mitigation strategies

Mark 6: Balanced evaluation

COMMON MISTAKES

- No human link made
- Only consequences, no responses

EXAMINER TIP

Evaluate = evidence + arguments + judgement.

GRADE 7–9 EXTENSION

Discuss correlation vs causation and climate models.

QUESTION 116

A power station releases 440 kg of carbon dioxide in one hour. Express this as grams per minute.

QUESTION TYPE

Topic: Chemistry of the atmosphere | **Command word:** calculate | **Marks:** 3 | **Skill tested:** Calculation

WHAT THE QUESTION IS ASKING

Convert kg to g and divide by 60 minutes.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

440 kg = 440,000 g. Rate = $440,000 \div 60 = 7333 \text{ g/min}$ (or $7.3 \times 10^3 \text{ g/min}$).

MARK BREAKDOWN

Mark 1: Convert 440 kg to 440,000 g

Mark 2: Divide by 60

Mark 3: 7333 g/min

COMMON MISTAKES

- Not converting kg to g
- Dividing by 3600 only

EXAMINER TIP

Check unit conversions step by step.

QUESTION 117

Describe an experiment to investigate the products of burning a hydrocarbon wax in plenty of air and in limited air.

QUESTION TYPE

Topic: Chemistry of the atmosphere | **Command word:** describe | **Marks:** 4 | **Skill tested:** Combustion practical

WHAT THE QUESTION IS ASKING

Compare complete vs incomplete combustion.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Burn wax on a balance in open air — mass gain from oxygen uptake, test gases with limewater (CO_2) and cobalt chloride paper (H_2O). Repeat with a lid restricting air — soot forms, CO may be present (test with blood-red complex with haemoglobin simulation or stated hazard).

MARK BREAKDOWN

- Mark 1: Plenty of air: CO_2 and H_2O products
- Mark 2: Limited air: soot/C produced
- Mark 3: Gas tests described
- Mark 4: Compare observations

COMMON MISTAKES

- No distinction between conditions
- Wrong gas tests

EXAMINER TIP

Contrast products under different oxygen supply.

Section 2: Using resources

QUESTION 118

State one advantage of recycling metals.

QUESTION TYPE

Topic: Using resources | Command word: state | Marks: 1 | Skill tested: Recycling

WHAT THE QUESTION IS ASKING

Saves energy, ore, landfill space.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Uses less energy than extracting from ore (or conserves finite ores / reduces landfill).

MARK BREAKDOWN

Mark 1: Uses less energy than extracting from ore (or conserves finite ores / reduces landfill).

COMMON MISTAKES

- Increases ore use
- Makes metal impure only

EXAMINER TIP

Energy saving is the classic answer.

QUESTION 119

Give the name of the process that separates crude oil into fractions.

QUESTION TYPE

Topic: Using resources | Command word: give | Marks: 1 | Skill tested: Crude oil

WHAT THE QUESTION IS ASKING

Fractional distillation.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Fractional distillation.

MARK BREAKDOWN

Mark 1: Fractional distillation.

COMMON MISTAKES

- Cracking only
- Filtration

EXAMINER TIP

Separation $\neq$ cracking.

QUESTION 120

State why potable water is not the same as pure water.

QUESTION TYPE

Topic: Using resources | Command word: state | Marks: 1 | Skill tested: Potable water

WHAT THE QUESTION IS ASKING

Safe to drink but contains dissolved substances.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Potable water is safe to drink but may still contain dissolved substances at low, safe levels.

MARK BREAKDOWN

Mark 1: Potable water is safe to drink but may still contain dissolved substances at low, safe levels.

COMMON MISTAKES

- Identical to distilled water
- Contains harmful bacteria

EXAMINER TIP

Potable = safe; pure = only H_2O molecules.

QUESTION 121

Describe how copper can be extracted from low-grade ores using phytomining.

QUESTION TYPE

Topic: Using resources | Command word: describe | Marks: 2 | Skill tested: Alternative extraction

WHAT THE QUESTION IS ASKING

Plants absorb metals, harvest and burn, extract copper.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Plants are grown on low-grade ore soil and absorb copper compounds. Plants are harvested, burned, and ash leached to extract copper ions, which are converted to copper metal.

MARK BREAKDOWN

Mark 1: Plants absorb metal ions from soil

Mark 2: Harvest, burn and leach ash to obtain copper

COMMON MISTAKES

- Uses bacteria in heaps directly for copper
- Plants create copper from nothing

EXAMINER TIP

Sequence: grow → harvest → burn → leach.

QUESTION 122

Explain why aluminium cannot be extracted from its ore by heating with carbon.

QUESTION TYPE

Topic: Using resources | Command word: explain | Marks: 2 | Skill tested: Extraction of metals

WHAT THE QUESTION IS ASKING

Al more reactive than carbon.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Aluminium is more reactive than carbon, so carbon cannot displace aluminium from aluminium oxide by reduction.

MARK BREAKDOWN

Mark 1: Aluminium more reactive than carbon

Mark 2: Electrolysis needed instead

COMMON MISTAKES

- Carbon is too expensive
- Aluminium ore is a liquid

EXAMINER TIP

Reactivity series determines extraction method.

QUESTION 123

Give two stages in treating fresh water to make it potable.

QUESTION TYPE

Topic: Using resources | Command word: give | Marks: 2 | Skill tested: Water treatment

WHAT THE QUESTION IS ASKING

Screening, sedimentation, filtration, chlorination.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Screening to remove large debris, and chlorination to kill harmful microbes.

MARK BREAKDOWN

Mark 1: Screening/filtration

Mark 2: Chlorination/sterilisation

COMMON MISTAKES

- Fractional distillation of seawater at all plants
- Adding salt

EXAMINER TIP

Any two valid treatment stages score.

QUESTION 124

Describe the process of bioleaching to extract copper from ores.

QUESTION TYPE

Topic: Using resources | **Command word:** describe | **Marks:** 3 | **Skill tested:** Bioleaching

WHAT THE QUESTION IS ASKING

Bacteria produce leachate with metal ions.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Bacteria are grown on low-grade ore heaps. They produce acidic leachate that dissolves copper compounds. The copper solution is collected and electrolysed or displaced to obtain copper metal.

MARK BREAKDOWN

- Mark 1: Bacteria on ore heaps
- Mark 2: Produce acidic solution dissolving copper
- Mark 3: Copper recovered from solution

COMMON MISTAKES

- Plants absorb copper
- High temperature smelting

EXAMINER TIP

Bioleaching uses bacteria, not plants.

QUESTION 125

Explain why life-cycle assessment (LCA) is useful when choosing a material for a product.

QUESTION TYPE

Topic: Using resources | **Command word:** explain | **Marks:** 3 | **Skill tested:** Life-cycle assessment

WHAT THE QUESTION IS ASKING

Considers all stages from raw material to disposal.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

LCA evaluates environmental impact at every stage — extracting raw materials, manufacturing, use, transport, and disposal — so choices are not based on one stage alone.

MARK BREAKDOWN

Mark 1: All stages considered

Mark 2: Avoids misleading single-stage comparison

Mark 3: Informs sustainable choice

COMMON MISTAKES

- Only manufacturing considered
- LCA ignores disposal

EXAMINER TIP

LCA is cradle-to-grave analysis.

QUESTION 126

Compare extracting iron from iron ore with recycling scrap iron for making new steel.

QUESTION TYPE

Topic: Using resources | **Command word:** compare | **Marks:** 4 | **Skill tested:** Recycling metals

WHAT THE QUESTION IS ASKING

Energy, ore use, quality considerations.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Extraction from ore uses blast furnace, high energy, raw materials, and produces CO₂. Recycling scrap uses less energy, conserves ore, reduces landfill, but sorting and melting still have costs; quality may require blending.

MARK BREAKDOWN

Mark 1: Extraction: high energy, CO₂, finite ore

Mark 2: Recycling: less energy, saves ore

Mark 3: Both have economic and environmental factors

Mark 4: Recycling not always 100% efficient

COMMON MISTAKES

- Recycling uses more energy
- No environmental comparison

EXAMINER TIP

Compare energy, resources and emissions.

QUESTION 127

Describe how seawater can be made potable by desalination and give one disadvantage.

QUESTION TYPE

Topic: Using resources | **Command word:** describe | **Marks:** 4 | **Skill tested:** Desalination

WHAT THE QUESTION IS ASKING

Distillation or reverse osmosis; energy cost.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.

2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Seawater is heated to distil fresh water (or forced through membranes by reverse osmosis), leaving salts behind. Disadvantage: high energy cost and therefore expense and carbon footprint.

MARK BREAKDOWN

- Mark 1: Distillation or reverse osmosis described
- Mark 2: Salt removed to produce fresh water
- Mark 3: Disadvantage: high energy/cost
- Mark 4: Environmental or economic disadvantage explained

COMMON MISTAKES

- Simple filtration removes salt
- No disadvantage given

EXAMINER TIP

Name process and energy disadvantage.

QUESTION 128

Evaluate the use of bioleaching and phytomining compared with traditional mining for copper. Consider environmental, economic and social factors.

QUESTION TYPE

Topic: Using resources | Command word: evaluate | Marks: 6 | Skill tested: Sustainable extraction

WHAT THE QUESTION IS ASKING

Extended evaluation of alternative extraction methods.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Traditional mining: high energy, landscape damage, dust, but well established. Bioleaching: low-grade ores, less landscape disruption, slow, bacterial control needed. Phytomining: very low-grade land use, renewable plants, slow, small yield. Both alternatives reduce energy versus smelting but take longer and suit low-grade ores.

MARK BREAKDOWN

- Mark 1: Traditional mining impacts stated
- Mark 2: Bioleaching advantages/limitations
- Mark 3: Phytomining advantages/limitations
- Mark 4: Environmental comparison
- Mark 5: Economic/social factors
- Mark 6: Balanced conclusion

COMMON MISTAKES

- Only one method discussed
- No evaluation judgement

EXAMINER TIP

Evaluate = pros and cons of each with conclusion.

GRADE 7–9 EXTENSION

Discuss scalability and land use requirements.

QUESTION 129

Recycling 1 kg of aluminium saves 95% of the energy needed to extract 1 kg from ore. If extraction uses 200 MJ, calculate the energy saved by recycling 1 kg.

QUESTION TYPE

Topic: Using resources | **Command word:** calculate | **Marks:** 3 | **Skill tested:** Calculation

WHAT THE QUESTION IS ASKING

Find 95% of extraction energy.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Energy saved = $0.95 \times 200 = 190$ MJ.

MARK BREAKDOWN

- Mark 1: 95% of 200 MJ calculated
- Mark 2: 190 MJ
- Mark 3: Correct unit MJ

COMMON MISTAKES

- Using 5% instead of 95%
- Subtracting wrong values

EXAMINER TIP

Percentage saved = multiply by percentage as a decimal.

QUESTION 130

Describe how you would distil a sample of muddy water to obtain pure water.

QUESTION TYPE

Topic: Using resources | **Command word:** describe | **Marks:** 4 | **Skill tested:** Required practical — water purification

WHAT THE QUESTION IS ASKING

Heat, condense, collect distillate.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Heat muddy water in a flask. Steam passes into a condenser cooled by running water. Condensed pure water (distillate) is collected in a beaker. Residue remains in the flask.

MARK BREAKDOWN

Mark 1: Heat to boil water

Mark 2: Condenser cools steam

Mark 3: Distillate collected separately

Mark 4: Residue left behind

COMMON MISTAKES

- Filter alone produces pure water
- No condenser described

EXAMINER TIP

Distillation involves boiling and condensing.

Section 3: Calculation walkthroughs

Calculation 1: Relative formula mass

Calculate the M_r of sulfuric acid, H_2SO_4 . (A_r : H = 1, S = 32, O = 16)

Formula: M_r = sum of A_r values for all atoms in the formula

Rearrangement: Not required — substitute directly into the sum

Substitution: $M_r = (2 \times 1) + 32 + (4 \times 16)$

Working: $M_r = 2 + 32 + 64 = 98$

Answer: $M_r = 98$ (no units)

Bracket the oxygen term: 4×16 before adding.

Calculation 2: Moles = mass $\div$ M_r

How many moles are in 8.8 g of carbon dioxide, CO_2 ? (M_r of $CO_2 = 44$)

Formula: moles = mass $\div$ M_r

Rearrangement: moles = mass $\div$ M_r

Substitution: moles = $8.8 \div 44$

Working: moles = 0.20 mol

Answer: 0.20 mol

Include the unit mol in your final answer.

Calculation 3: Concentration

Calculate the concentration in mol/dm³ when 0.25 mol of NaCl is dissolved to make 500 cm³ of solution.

Formula: concentration (mol/dm³) = moles $\div$ volume (dm³)

Rearrangement: $c = n \div V$

Substitution: $V = 500 \div 1000 = 0.50$ dm³; $c = 0.25 \div 0.50$

Working: $c = 0.50$ mol/dm³

Answer: 0.50 mol/dm³

Always convert cm³ to dm³ by dividing by 1000.

Calculation 4: Reacting masses

What mass of magnesium oxide is produced when 1.2 g of magnesium burns completely? $2Mg + O_2 \rightarrow 2MgO$
(A_r : Mg = 24, O = 16)

Formula: moles = mass $\div$ A_r ; use molar ratio from equation; mass = moles $\times$ M_r

Rearrangement: mass MgO = moles Mg $\times$ M_r (MgO)

Substitution: Moles Mg = $1.2 \div 24 = 0.05$ mol. Ratio 2:2 so moles MgO = 0.05. M_r MgO = 40.

Working: Mass MgO = $0.05 \times 40 = 2.0$ g

Answer: 2.0 g

Check the balanced equation ratio before converting moles.

Calculation 5: Percentage yield

A student expected 10.0 g of copper but obtained 8.5 g. Calculate the percentage yield.

Formula: percentage yield = (actual yield ÷ theoretical yield) × 100

Rearrangement: % yield = (actual ÷ theoretical) × 100

Substitution: % yield = (8.5 ÷ 10.0) × 100

Working: % yield = 85%

Answer: 85%

Actual yield is always on top — it is what you really got.

Calculation 6: Atom economy

Calculate the atom economy for making hydrogen: $\text{CH}_4 + 2\text{H}_2\text{O} \rightarrow \text{CO}_2 + 4\text{H}_2$. (A_r: H = 1, C = 12, O = 16)

Formula: atom economy = (M_r of desired product × its coefficient) ÷ (total M_r of reactants) × 100

Rearrangement: Substitute into the percentage formula directly

Substitution: Desired: $4\text{H}_2 = 4 \times 2 = 8$. Reactants: $16 + 36 = 52$.

Working: Atom economy = $(8 \div 52) \times 100 = 15.4\%$

Answer: 15.4% (accept 15%)

Only the desired product mass goes on the numerator.

Calculation 7: Gas volumes

What volume, in dm³, does 0.5 mol of gas occupy at room temperature and pressure? (Molar gas volume = 24 dm³/mol)

Formula: volume = moles × molar gas volume

Rearrangement: $V = n \times 24$

Substitution: $V = 0.5 \times 24$

Working: $V = 12 \text{ dm}^3$

Answer: 12 dm³

At rtp, 1 mol of any gas occupies 24 dm³.

Calculation 8: Titration calculation

25.0 cm³ of 0.100 mol/dm³ NaOH neutralises 23.50 cm³ of HCl. Calculate the concentration of HCl. $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$

Formula: moles = concentration × volume (dm³); $c_1V_1 = c_2V_2$ for 1:1 reactions

Rearrangement: $c(\text{HCl}) = \text{moles HCl} \div \text{volume HCl (dm}^3\text{)}$

Substitution: Moles NaOH = $0.100 \times 0.0250 = 0.00250 \text{ mol}$. Moles HCl = 0.00250. $V \text{ HCl} = 0.02350 \text{ dm}^3$.

Working: $c(\text{HCl}) = 0.00250 \div 0.02350 = 0.106 \text{ mol/dm}^3$

Answer: 0.106 mol/dm³ (accept 0.11 mol/dm³)

Concordant titres should be averaged before calculating.

Section 4: Required practical walkthroughs

Practical 1: Acid–alkali titration (required practical)

Aim: To find the concentration of hydrochloric acid by titration with sodium hydroxide solution of known concentration.

Variables: Independent: acid added from burette. Dependent: volume of acid (titre). Control: fixed volume and concentration of alkali, same indicator, same technique.

Method: Rinse the pipette with sodium hydroxide solution and transfer 25.0 cm³ into a conical flask. Add 2–3 drops of methyl orange indicator. Fill a burette with hydrochloric acid and record the initial reading. Add acid slowly, swirling after each addition, until the indicator just changes colour. Record the final burette reading. Repeat until concordant titres (within 0.10 cm³) are obtained.

Risk assessment: Acids and alkalis are irritants — wear eye protection and rinse spills with water. Breakage of glassware — handle burette and pipette carefully.

Results table: Record initial and final burette readings and calculate titre = final – initial. Mean titre from concordant results. Example: titre = 23.6 cm³.

Graph guidance: Not usually required for titration; a results table of titre values is sufficient. Plot titre vs trial number to check consistency if desired.

Conclusion: Use moles = $c \times V$ for NaOH, 1:1 ratio for HCl, then $c(\text{HCl}) = \text{moles} \div \text{volume in dm}^3$. State the calculated concentration with units mol/dm³.

Evaluation: Sources of error: parallax when reading burette, overshooting endpoint, incomplete mixing. Improve by adding acid dropwise near endpoint and reading burette at eye level.

Common exam questions:

- Why is a pipette used for the alkali?
- Why are concordant titres important?
- How would you know you had reached the endpoint?

Model answers:

- A pipette delivers an accurate, fixed volume of alkali.
- Concordant results show the titration is reliable; the mean of concordant titres is used in calculations.
- The indicator changes colour permanently (e.g. methyl orange turns from yellow to red in excess acid).

Practical 2: Temperature change and rates of reaction (required practical)

Aim: To investigate how temperature affects the rate of reaction between sodium thiosulfate and hydrochloric acid.

Variables: Independent: temperature of thiosulfate solution. Dependent: time for cross to disappear. Control: concentrations, volumes, same cross, same flask.

Method: Draw a cross on paper and place under a conical flask. Heat sodium thiosulfate to a set temperature (water bath). Add 5 cm³ hydrochloric acid, start a timer and swirl. Stop when the cross is no longer visible. Repeat at different temperatures (e.g. 20, 30, 40, 50 °C). Record time for each temperature.

Risk assessment: Acids are irritants — wear goggles. Hot water baths — use tongs and avoid burns.

Results table: Table: Temperature (°C) | Time (s). Example: 20 °C → 80 s; 40 °C → 35 s; 60 °C → 15 s. Shorter time = faster rate.

Graph guidance: Plot temperature (x-axis) against time (y-axis) OR rate (1/time) against temperature. Rate increases with temperature — curve or line decreasing time.

Conclusion: As temperature increases, time decreases so rate increases. Higher temperature gives particles more kinetic energy and more collisions exceed activation energy.

Evaluation: Errors: judging exactly when cross disappears, temperature drop before mixing, inconsistent volumes. Improve with multiple repeats, thermometer in solution until mixing, and video replay for timing.

Common exam questions:

- Why is the cross used in this experiment?
- How does the graph show rate increases with temperature?
- State one control variable.

Model answers:

- The cross provides a visible endpoint — when sulfur precipitate obscures it, the reaction has progressed enough to compare times.
- Time decreases as temperature rises (or 1/time increases), showing faster rate at higher temperature.
- Same concentration of reactants / same volumes / same cross visibility distance.

Practical 3: Making salts by titration and crystallisation

Aim: To prepare a pure, dry sample of a soluble salt from an acid and an alkali.

Variables: Independent: neutralisation completion. Dependent: mass of pure dry salt. Control: same concentrations, same volumes after neutralisation.

Method: Titrate acid and alkali with indicator to find neutralisation volume, then repeat without indicator using the known volume. Heat the salt solution gently to evaporate some water, leave to crystallise, filter crystals, and dry between filter papers.

Risk assessment: Hot solutions and Bunsen flames — use tongs and heat gently. Acids and alkalis irritant.

Results table: Record mass of dry crystals obtained. Example: 2.8 g of sodium chloride crystals.

Graph guidance: Not required; record mass of product from each trial if repeating.

Conclusion: Neutralisation produces a salt and water. Evaporation and crystallisation give a pure solid salt.

Evaluation: Losses during transfer and incomplete crystallisation reduce yield. Improve by evaporating slowly and leaving longer to crystallise.

Common exam questions:

- Why is the titration done twice — once with indicator and once without?
- Why is the solution heated gently?

Model answers:

- First titration finds the exact neutralisation volume; second prepares the salt without contaminating it with indicator.
- Gentle heating evaporates water to saturation so crystals form without decomposing the salt or spitting hot solution.

Section 5: Full mini mock paper

Total marks: 50

Question 1 (1 marks)

State the charge on a neutron.

Question 2 (2 marks)

Give the electron arrangement in an atom of fluorine (atomic number 9).

Question 3 (3 marks)

Explain why ionic compounds have high melting points.

Question 4 (3 marks)

Calculate the M_r of nitric acid, HNO_3 . (A_r : H = 1, N = 14, O = 16)

Question 5 (2 marks)

Describe the test for chlorine gas.

Question 6 (4 marks)

A student adds zinc to copper(II) sulfate solution. Describe what you would observe and explain the reaction.

Question 7 (3 marks)

Explain, using collision theory, why a catalyst increases the rate of reaction.

Question 8 (3 marks)

Draw conclusions from this data: at 20 °C the reaction time is 60 s; at 40 °C it is 25 s. What does this show about temperature and rate?

Question 9 (4 marks)

Describe how crude oil is separated into fractions.

Question 10 (3 marks)

Calculate the percentage yield if theoretical yield is 15.0 g and actual yield is 12.0 g.

Question 11 (4 marks)

Explain how greenhouse gases contribute to global warming.

Question 12 (6 marks)

Evaluate the advantages and disadvantages of recycling aluminium.

Question 13 (4 marks)

A titration uses 25.0 cm³ of 0.0800 mol/dm³ NaOH to neutralise 22.10 cm³ of sulfuric acid. $\text{H}_2\text{SO}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$. Calculate the concentration of sulfuric acid in mol/dm³.

Question 14 (6 marks)

Describe how paper chromatography can identify dyes in a mixture.

Question 15 (2 marks)

State two environmental problems caused by burning fossil fuels.

Mini mock — step-by-step walkthrough answers

Answer 1

A neutron has no charge — it is neutral (0).

Mark breakdown:

Mark 1: Neutron charge is 0 / neutral

Answer 2

Fluorine has 9 electrons: arrangement 2,7 (two in inner shell, seven in outer shell).

Mark breakdown:

Mark 1: 2 electrons in first shell

Mark 2: 7 electrons in second shell

Answer 3

Ionic compounds have giant ionic lattices with strong electrostatic forces between oppositely charged ions. A large amount of energy is needed to break these forces, so melting points are high.

Mark breakdown:

Mark 1: Giant ionic lattice

Mark 2: Strong electrostatic forces between ions

Mark 3: Much energy needed to break lattice

Answer 4

$M_{\text{r}} = 1 + 14 + (3 \times 16) = 1 + 14 + 48 = 63$.

Mark breakdown:

Mark 1: Correct substitution

Mark 2: $M_{\text{r}} = 63$

Mark 3: No units for M_{r}

Answer 5

Hold damp blue litmus paper at the mouth of the tube — it bleaches white. (Accept: green gas, choking smell as observation supporting identity.)

Mark breakdown:

Mark 1: Damp litmus paper bleaches

Mark 2: Identifies chlorine

Answer 6

The blue solution fades, a brown coating of copper forms on the zinc, and the zinc diminishes. Zinc is more reactive than copper, so it displaces copper from solution: $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$.

Mark breakdown:

Mark 1: Blue colour fades

Mark 2: Brown copper on zinc

Mark 3: Zinc more reactive than copper

Mark 4: Displacement reaction explained

Answer 7

A catalyst provides an alternative pathway with lower activation energy, so a greater proportion of collisions are successful, increasing rate without being used up.

Mark breakdown:

- Mark 1: Alternative pathway
- Mark 2: Lower activation energy
- Mark 3: More successful collisions

Answer 8

Time halves (approximately) when temperature rises 20 °C, so rate increases with temperature. Higher temperature gives faster particle movement and more collisions exceeding activation energy.

Mark breakdown:

- Mark 1: Shorter time at higher temperature
- Mark 2: Rate increases with temperature
- Mark 3: Linked to collisions / activation energy

Answer 9

Crude oil is heated and vapourised. It enters a fractionating column where fractions condense at different heights depending on boiling point — gases at the top, heavy bitumen at the bottom.

Mark breakdown:

- Mark 1: Crude oil heated/vapourised
- Mark 2: Fractionating column
- Mark 3: Separated by boiling point
- Mark 4: Named fractions or top-to-bottom order

Answer 10

% yield = $(12.0 \div 15.0) \times 100 = 80\%$.

Mark breakdown:

- Mark 1: Correct formula used
- Mark 2: $12.0 \div 15.0$
- Mark 3: 80%

Answer 11

Greenhouse gases absorb infrared radiation emitted from Earth and re-emit it, reducing heat loss to space. This enhanced greenhouse effect raises average global temperatures.

Mark breakdown:

- Mark 1: Absorb infrared radiation
- Mark 2: Trap heat / reduce energy loss to space
- Mark 3: Leads to global temperature rise
- Mark 4: Linked to climate change

Answer 12

Advantages: saves energy (up to 95% vs extraction), conserves finite bauxite ore, reduces landfill and mining damage. Disadvantages: collection and sorting costs, transport emissions, not all aluminium is recycled. Overall recycling is environmentally beneficial where infrastructure exists.

Mark breakdown:

- Mark 1: Energy saving advantage
- Mark 2: Conserves ore / reduces mining
- Mark 3: Disadvantage: cost or sorting
- Mark 4: Disadvantage: transport or not 100% recycled
- Mark 5: Environmental evaluation
- Mark 6: Balanced judgement

Answer 13

Moles NaOH = $0.0800 \times 0.0250 = 0.00200$ mol. Ratio 1:2 so moles $\text{H}_2\text{SO}_4 = 0.00100$ mol. Volume = 0.02210 dm^3 . $c = 0.00100 \div 0.02210 = 0.0452 \text{ mol/dm}^3$.

Mark breakdown:

- Mark 1: Moles NaOH = 0.00200
- Mark 2: Moles $\text{H}_2\text{SO}_4 = 0.00100$ (1:2 ratio)
- Mark 3: Volume converted to dm^3
- Mark 4: $c = 0.0452 \text{ mol/dm}^3$

Answer 14

Spot sample on pencil baseline, place in solvent below the line, allow separation. Measure distances and calculate $R_f = \text{distance by dye} \div \text{distance by solvent}$. Compare with known R_f values to identify dyes. Explain why R_f is constant for a given dye/solvent pair.

Mark breakdown:

- Mark 1: Baseline and solvent method
- Mark 2: Dyes separate on paper
- Mark 3: R_f calculated correctly
- Mark 4: Comparison with standards
- Mark 5: R_f defined or explained
- Mark 6: Clear identification conclusion

Answer 15

Any two from: climate change (CO_2 greenhouse effect), acid rain (SO_2/NO_2), air pollution / respiratory problems, production of particulates.

Mark breakdown:

- Mark 1: First valid problem
- Mark 2: Second valid problem